Multiplexed CRISPRi Perturb-seq in Yeast

Methods, Theoretical Limits, and Genome-Scale Cost

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Abstract

Pooled genetic screens read out by single-cell RNA sequencing map perturbations to their transcriptional consequences at dense, transcriptome-wide resolution. This work establishes what such a screen costs in *S. cerevisiae*, what sets the ceiling on how many genes can be perturbed in one cell, and which parameter is binding. It serves two purposes: selecting a method, and explaining the chemistry and the statistics well enough that the relevant passages can be reused in a methods write-up. Numbers are regenerated from committed scripts rather than transcribed, and every claim not backed by a hash-pinned artifact in the library mirror carries a visible flag, listed in the key below.

Working notes: `notes/microbe-perturb-seq.*.md`

Generating scripts: `experiments/019-perturb-seq-costing/`

References: Zotero `torchcell/microbe-perturb-seq` (group ∪ personal)

✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

▲ number not in our library mirror; verify at the linked source ● taken from a review's summary table, not the primary paper

Contents

1 Introduction ✗

Metabolic engineering builds strains by stacking perturbations, but the design decisions are made from data on single perturbations measured one at a time. Pooled CRISPR interference screens read out by single-cell RNA sequencing (**Perturb-seq**) collapse that gap: a library of guide RNAs is transformed into a population, every cell is sequenced, and each cell's guide identity is recovered from its own transcriptome, so one experiment yields a matrix from perturbation to full transcriptional response rather than a list of fitness hits [1]. It is the highest-density source of genotype-to-expression training data available. A growth screen scores only perturbations that change fitness; a transcriptional readout also scores those that are fitness-neutral but transcriptionally loud — though that advantage is a property of the *single-cell* measurement, and it weakens to the extent that a design falls back on clonal populations to accumulate enough transcripts per genotype (section 4.3).

Three developments have made the yeast version tractable. A split-pool barcoding chemistry removes the instrument from the cell-isolation step [2]; pooled CRISPRi and single-cell transcriptomics have been joined in bacteria, the first demonstration in any microbe [3]; and a genome-scale yeast deletion collection has been rebuilt so its genotype tag is transcribed and therefore visible to the assay [4]. What follows establishes what such a screen costs in *S. cerevisiae*, how many genes can be perturbed simultaneously before the measurement stops being interpretable, and which parameter is actually binding — because the answer is not the one the published per-cell costs suggest.

2 Measurement Principles and Terminology ✗

This section fixes the vocabulary. Everything after it assumes these terms.

2.1 Molecular Identifiers ✗

Table 1 is the reference: every term the document uses in a technical sense, with a link to the section that treats it. It is worth a first pass rather than only being consulted, because the naming is genuinely treacherous — four distinct things in this document are called a "barcode", two different things are called an "index", and the difference between cells per *target gene* and cells per *guide* is a factor of six that changes every budget in section 5. The prose below then develops the four identifiers properly.

Table 1: Controlled vocabulary, alphabetical. The last column links to the section that treats each term in depth, so an entry is a pointer rather than a replacement for the argument. Generated from `experiments/019-perturb-seq-costing/scripts/glossary.py`.

Term	Definition	See
Barcode collision	Two distinct cells receiving the same full barcode and being merged into one apparent cell. Rate is $1-((B-1)/B)^{C-1}$ for C cells in a barcode space of size B, so it is controlled by adding rounds or sublibraries, not by sequencing harder.	section 5.4
Biological floor	The minimum cells a perturbation needs before its mean is interpretable, independent of sequencing. Cell-cycle phase and metabolic state must be averaged out, and depth cannot substitute. The published field standard is 100 cells.	section 4.3
Biological overdispersion (φ_j)	Variance in a gene's expression between genetically identical cells in excess of shot noise, from cell-cycle phase, metabolic state and transcriptional bursting. Reading each cell more deeply does not reduce it; only more cells do. Equal to the negative-binomial dispersion, so $\sqrt{\varphi_j}$ is the biological coefficient of variation reported in the RNA-seq literature.	section 4.7
Blocking strand	Added after each ligation round to outcompete the linker and retire that well's barcode, so it cannot participate in a later round. Without it, leftover oligos scramble the barcode sequence.	section 3.1
Cell barcode (CB)	A sequence shared by every cDNA molecule from one cell and ideally unique to it; what makes the data single-cell rather than bulk. Delivered pre-synthesised on a bead in droplet methods, built up across rounds of well-specific ligation in split-pool.	section 2.1
Cells per target gene	Cells carrying a guide against that gene, pooled over all of its guides. *Not* cells per guide and *not* cells per plasmid – the distinction is a factor of six here and is the most common way a screen budget is misread.	section 4.3
Channel	One microfluidic lane of a droplet chip, and the unit droplet reagent is sold in. A Chromium chip runs several channels side by side; each is loaded and priced independently and recovers up to ∼20,000 cells, so more cells means proportionally more channels and proportionally more money. That is what "cost scales with cells" means concretely. A split-pool plate has no equivalent – it is priced per plate however many cells pass through it.	section 3.2

Table 1, continued

Term	Definition	See
Combinatorial split-pool barcoding	No isolation at all. Fixed, permeabilised cells act as their own reaction vessels; the population is split across a plate, barcoded in situ, pooled, and re-split. After R rounds of 96 wells there are 96^R possible paths. Cost scales per well, not per cell, so a plate costs the same for ten thousand cells or a million.	section 2.2
Depth sufficiency (d^*)	The per-cell depth $1/(\varphi_j p_j)$ at which shot noise and biological overdispersion contribute equally. Below it a screen is measurement-limited and deeper reads help; above it the requirement is within a factor of two of a floor no sequencing can cross.	section 4.7
Doublet	Two cells reported as one. In droplet methods it is two cells in one partition, governed by Poisson loading (10x quote ∼8% at 20,000 cells recovered); in split-pool the analogous failure is a barcode collision, which has a different and controllable cause.	section 2.2
Droplet isolation	One cell co-encapsulated with one barcoded bead in an emulsion droplet; the droplet wall is what keeps a cell's molecules together. Reagent cost scales linearly with cells, because you pay per channel.	section 2.2
FR-Perturb	The decoder for a compressed screen: factorises the observed expression matrix by sparse PCA, then recovers per-perturbation effects by LASSO against the pooling design. The recovered effects are estimates, so power depends on the design matrix as much as on cell count.	section 3.3
Guide barcode (GBC)	An expressed, polyadenylated sequence that stands in for the guide RNA itself. A Pol III guide cassette carries no poly(A) tail and is therefore invisible to 3′ capture, so the perturbation identity has to be re-encoded in something the assay can see.	section 3.5
Guide-pooling vs cell-pooling	Two ways to compress a screen. Guide-pooling puts several guides in one cell; cell-pooling combines separately perturbed cells before sequencing. Both rely on the perturbations composing additively, and both are decoded rather than read directly.	section 3.3
Linker strand	A splint oligo pre-annealed to a barcode, positioning it against the 5′ end of the growing strand so T4 DNA ligase can seal the nick. Ligation is templated by the linker, which is why a stray barcode from an earlier well would corrupt the address.	section 3.1
Main effect	The average effect of perturbing one gene, pooled over every cell carrying a guide against it regardless of what else that cell carries. Well defined only under an additive (log-additive) model.	section 4.5
Perturb-seq	A pooled genetic screen read out by single-cell RNA-seq, so each cell yields both a perturbation identity and a whole-transcriptome phenotype. The transcriptome, not a growth rate, is the phenotype.	section 2
PhiX spike-in	A fraction of the loaded library that is control phage DNA. At ≥10% it is a split-pool protocol requirement rather than hygiene: read 2 carries fixed linker sequence at defined cycles, so base diversity collapses and patterned flow cells mis-register clusters without it. Budget it as a tax on usable reads.	section 5.3
Plex (k)	Guide RNAs delivered per cell, so k genes are perturbed in that cell at once. Raising k divides the cell requirement for main effects by k; it does not make genome-wide pairwise interactions reachable.	section 4.5
Protocol run	One complete pass through the barcoding workflow – fix, permeabilise, three rounds, split into sublibraries. Consumes one withdrawal from each barcode plate and carries up to ∼480,000 cells. A unit of *bench work*, not of sequencing: the sublibraries it yields are sequenced and priced separately.	section 5.1
Pseudobulk	Summing UMIs across all cells sharing a perturbation, to estimate that perturbation's mean expression. It is what the whole design is powered on: single cells are too shallow individually, so the quantity that matters is cells × depth.	section 4.3
Read pair	One paired-end fragment, the unit sequencing is priced in here. Both method families need paired-end runs – one read carries the cDNA and the other the barcodes – but they put the barcode on opposite reads.	section 5.3
rRNA depletion	Removal of ribosomal cDNA from the amplified library, after barcoding and before fragmentation and indexing. It does not improve capture; it stops ∼94% of purchased reads being spent on rRNA, and is the single highest-leverage change available to a split-pool screen.	section 3.4
Semipermeable capsule (SPC)	A hydrogel shell, pore cutoff around 30 nm, that confines cells and their nucleic acids while letting enzymes diffuse in. A fourth isolation principle alongside droplet, plate and split-pool: like a droplet it is a physical container, but unlike a droplet it survives buffer exchange, so a capsule can be washed, sorted and carried through multi-step chemistry. Small molecules leak, which is what rules it out for a metabolomic readout.	section 6.3
Shot noise	The irreducible randomness of counting. Sequencing samples molecules at random, so a gene at true rate μ is observed with variance μ and relative error $1/\sqrt{\mu}$. A property of the measurement, not of the cell – and the only noise that buying more reads removes. Also called counting or sampling noise.	section 4

Table 1, continued

Term	Definition	See
Source plate / working plate	A source plate is the pre-diluted barcode-oligo stock as shipped; working plates are aliquots cycled in daily use. The distinction is economic: source plates are retired by freeze–thaw damage rather than by depletion, so they behave as a capital asset.	section 5.1
Spheroplast	A yeast cell whose β-glucan wall has been digested away, here with zymolyase under osmotic support. Required because no commercial chemistry can get reagents through an intact fungal wall; the trade is that a wall-less cell bursts in a normal buffer.	section 2.3
Splint	The same molecule as the linker strand, named for what it does: bridge two DNA ends so ligase can seal the nick between them. Ligation, not polymerization – which is the whole difference from a template-switching oligo, where a polymerase copies the oligo rather than a ligase joining to it. Split-pool uses splints; droplet $5'$ chemistry uses template switching.	section 3.1
Sublibrary	An aliquot of barcoded cells (5,000–20,000) split off after the last in-cell round, lysed and prepared as one Illumina library. Sublibrary count is set jointly by barcode-collision targets and by PCR complexity limits.	section 5.1
Sublibrary index	The Illumina index read added at final PCR. In split-pool it is not merely a sample tag: cells are divided into sublibraries *after* the last in-cell barcoding round, so the index is a further barcode dimension and two cells that took identical plate paths are still resolved if they landed in different sublibraries. This is why unique dual indices are mandatory – an index hop manufactures a barcode collision.	section 5.4
Tagmentation	Fragmentation and adapter addition in one step, performed by the Tn5 transposase. Tn5 is loaded with adapter duplexes, cuts double-stranded DNA and ligates an adapter to each cut end in the same reaction, so it replaces a separate shear-then-ligate workflow. Figures draw it as an enzyme body rather than a sequence block, because it is a protein acting on the molecule, not a piece of the molecule.	section 2.1
Template switch	A bead-borne reaction adding a common handle to the far end of the cDNA, giving the molecule PCR primer sites at both ends. Everything after it is conventional Illumina library prep.	section 3.1
Template-switching oligo (TSO)	The oligo that adds the second PCR handle in $5'$ chemistries, and *not* the same mechanism as a linker. Reverse transcriptase, on reaching the $5'$ end of an mRNA, adds two or three non-templated C residues; the TSO ends in rGrGrG, base-pairs with them, and the enzyme switches template and copies the TSO onto the cDNA. The handle is therefore written by the polymerase starting from the RNA's own $5'$ end, which is why a TSO-delivered barcode marks the $5'$ end of the transcript while an oligo-dT-delivered one marks the $3'$ end.	section 2.1
UMI collision	Two distinct molecules of the same gene in the same cell drawing the same UMI, and being counted once. Negligible at yeast expression levels: a 10 nt UMI has $4^{10} \approx 10^6$ states against at most a few thousand molecules of any one gene. Not to be confused with barcode collision, which is a cell-level error.	section 2.1
Unique molecular identifier (UMI)	A short random sequence attached to each cDNA molecule *before* amplification. Reads sharing CB, gene and UMI are PCR copies of one original mRNA, so collapsing on the UMI turns read counts into molecule counts. 10 nt in the split-pool protocols here, 10–12 nt in droplet.	section 2.1
Usable vs sequenced cell	A usable cell survives QC *and* carries an identified perturbation; a sequenced cell is one whose reads were paid for. The two differ by $\sim 4\times$ in split-pool, which is why a quoted cost per cell understates the real one.	section 5.2

A single-cell RNA-seq read is only useful if you can answer three questions about it: which cell did it come from, was it a distinct original molecule or a PCR copy, and which gene is it. Different chemistries answer these differently, but all of them attach the same four classes of tag.

Cell barcode (CB). A sequence shared by every cDNA molecule from one cell and, ideally, unique to that cell. It is what makes the data *single-cell* rather than bulk. In droplet methods the CB is delivered pre-synthesised on a bead; in split-pool methods it is *built up* across several rounds of well-specific ligation, so the barcode records the trajectory a cell took through a series of plates (Sec. 3.1).

Unique molecular identifier (UMI). A short random sequence — 10–12 nt in the methods here — added to each cDNA molecule *before* any amplification. Two reads sharing a CB, a gene, and a UMI are copies of one original mRNA; two reads sharing CB and gene but differing in UMI are two molecules. Collapsing on the UMI converts read counts, which are distorted by PCR efficiency, into molecule counts, which are the actual measurement. The count of distinct UMIs recovered from a cell is the single most important quality number in this document: it is the denominator of every statistical argument in Sec. 4. **UMI collision** — two distinct molecules of the same gene drawing the same UMI — is negligible at yeast expression levels, since a 10 nt UMI has $4^{10} \approx 10^6$ states against at most a few thousand molecules of any one gene.

Sample or sublibrary index. An Illumina index read added during the final PCR. In a conventional experiment it separates samples sharing a flow cell. In split-pool it does more than that, and the difference matters: because cells are divided into sublibraries *after* the last in-cell barcoding round, the index is effectively an additional barcode dimension, so two cells that took identical paths through the plates are still resolved if they landed in different sublibraries (Sec. 5.4).

Perturbation identifier. What makes the experiment a screen rather than a survey. This is the hard part in yeast and is treated separately in Sec. 3.5; the short version is that the guide RNA itself is usually invisible to the assay, so a proxy has to be engineered.

2.2 Cell Isolation Strategies ✗

Physical isolation (droplet). A microfluidic device co-encapsulates one cell with one barcoded bead in an emulsion droplet, which isolates every molecule that cell releases. Cell loading follows Poisson statistics, so a low occupancy is used to keep **doublets** — two cells in one droplet, reported as one — acceptably rare; 10x quote about 8% at 20,000 cells recovered.

Cost scales with cells, and the unit is worth being concrete about because it drives every droplet figure in section 5. Reagent is sold per **channel**: a Chromium chip carries several channels, each loaded and priced separately, and each recovers up to ∼20,000 cells. Two 20,000-cell recoveries are therefore two channels and twice the reagent cost — there is no per-cell dial, only a count of channels. At the UIUC rate a genome-scale screen at 250 cells per target gene needs 137 channels at $2,143 each (section 5.5), which is why library preparation, not sequencing, dominates the droplet column of the budget.

Combinatorial split-pool barcoding. No isolation at all. Cells are fixed and permeabilized so they become their own reaction vessels: their membranes retain the cDNA while enzymes diffuse in. The population is distributed across a 96-well plate, a well-specific barcode is attached in situ, the cells are pooled, redistributed, and barcoded again. After R rounds a cell carries an ordered barcode $\mathrm{BC}_1 \| \ldots \| \mathrm{BC}_R$, and with 96 wells per round there are 96^R possible paths. Cells sharing round 1 are overwhelmingly likely to differ by round 3.

The cost consequence is the important one: *you pay per well, not per cell.* A round of barcoding consumes one 96-well plate of oligo and one tube of enzyme whether ten thousand cells or a million pass through it, so the reagent line is flat in cell number and the only per-cell costs left are sequencing and the per-sublibrary prep. Brettner et al.'s barcoding reagent works out to $0.005 per cell and Gaisser et al.'s per-run barcode cost is flat by construction (section 5.1); the entire economic case for split-pool is this one line, and section 5.2 tests whether it survives contact with rRNA waste and discarded cells.

The landscape between the two poles. Those are the extremes; fig. 1 places the six approaches actually used in yeast between them and pairs each with the library chemistry it can support. The organizing fact is that the isolation choice and the barcode choice are not independent. A method that pools cells before sequencing has to write identity into the molecule, and *which* oligo carries the barcode fixes *where on the transcript* it ends up — an oligo-dT primer puts it at the 3′ end, a **template-switching oligo** at the 5′ end, Tn5 wherever it cuts. Plate methods keep one cell per well until the very end and need no cell barcode at all: the Illumina sample index identifies the well and the well identifies the cell. That index is a sample tag, not a **unique molecular identifier** — it names which library a read came from, whereas a UMI distinguishes two molecules *within* one cell, and a plate method still needs its own UMI to do that.

2.3 Microbial Constraints on Single-Cell Chemistry ✗

Every commercial single-cell chemistry was developed on mammalian cells, and three properties of a yeast cell break its assumptions.

- **A cell wall.** Reagents cannot diffuse in and the cell cannot lyse on the mammalian schedule. Both method families solve this with zymolyase, a β-glucanase. Digestion is not optional — without it no reagent reaches the transcriptome — but *when* it happens decides whether the cell survives to be isolated, because a wall-less spheroplast is fragile and lyses in a normal buffer. That timing is the whole of Jariani et al.'s contribution (Sec. 3.2).
- **Roughly a fifth the mRNA.** A yeast cell holds on the order of 10^4 mRNA molecules against 10^5 for a mammalian cell (Sec. 4.1). Less input means fewer UMIs recovered, which is why per-cell yeast data always looks poor next to a mammalian benchmark. It is also why cells are pooled rather than read individually: at a few hundred UMIs a single cell cannot estimate any gene's expression, but the cells carrying one perturbation are biological replicates of the same genotype, so summing their counts recovers that genotype's mean profile. The single-cell step is doing *assignment* — which cell carries which perturbation — and the pooling is doing the measurement (Sec. 2.4).

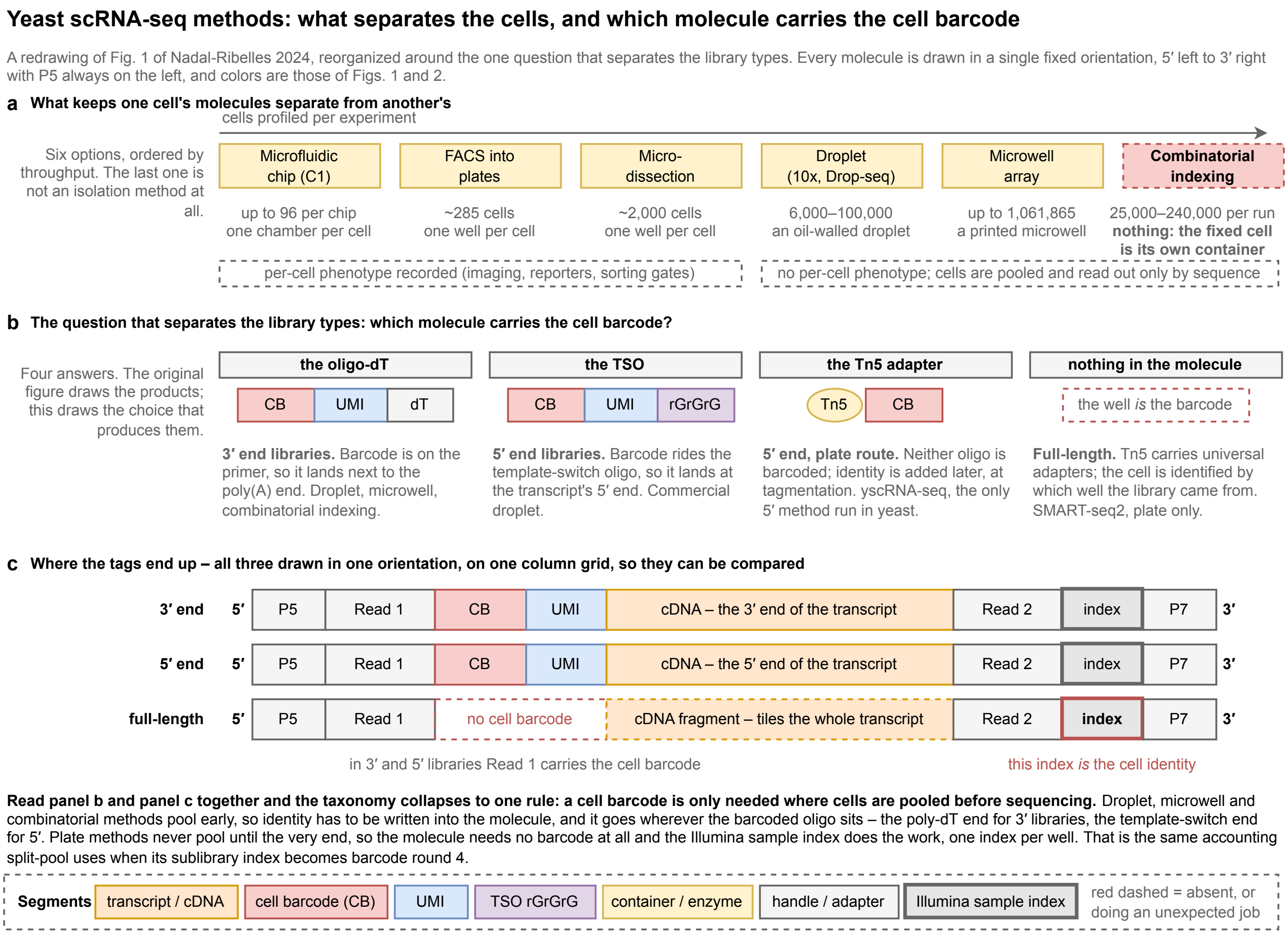

Figure 1: A cell barcode is only needed where cells are pooled before sequencing, and most of the method landscape follows from that one fact. The yeast scRNA-seq landscape, reorganized from the review of [5]. **(a)** What keeps one cell's molecules separate, ordered by throughput and each labeled with the container that does the work. Combinatorial indexing is set apart because it is not an isolation method at all: the fixed cell is its own vessel. The two dashed boxes beneath the row split the six by whether a per-cell phenotype is recorded, which is what decides whether a method can be paired with a reporter or a sorting gate. **(b)** The question that actually separates the library types, which is which molecule carries the cell barcode. There are four answers: the oligo-dT gives 3′ libraries, the template-switch oligo gives 5′, a barcode loaded into Tn5 gives the plate route used by the only 5′ method yet run in yeast, and in full-length libraries nothing in the molecule carries it. **(c)** Where the tags end up, with all three library types drawn on one column grid in one orientation so they can be compared: 3′ and 5′ differ only in which end of the transcript sits beside the barcode, while full-length carries no barcode anywhere and is resolved instead by the sample index, one per well — the same accounting that makes split-pool's sublibrary index barcode round 4 (Sec. 5.4). Redrawn and reorganized from the text and Table 1 of that review rather than reproduced from its figure, and with the orientation fixed 5′ to 3′ throughout, where the original mirrors its 3′ panel. Every number drawn by hand in this figure is sourced in table 3.

- **An overwhelming rRNA background.** About 85% of a yeast cell's RNA is ribosomal. This is a property of the RNA pool, not an unavoidable property of the library: ribosomal RNA is not polyadenylated, so an oligo-dT primer does not prime on it and no separate depletion step is needed. A strictly poly(A)-selective chemistry — 10x's bead oligo, for instance — therefore never sees most of that 85%. The problem arises only when a protocol deliberately adds a non-selective primer. Brettner et al. prime with oligo-dT *and* random hexamers, and hexamers anneal to anything including rRNA, which is why their recovered transcripts are 93.7% rRNA while Jariani et al.'s oligo-dT-only droplet run is not (Sec. 3.4). The hexamer is there to reach transcripts without a poly(A) tail, which is what carries the method to bacteria; in yeast it is largely a tax, and as Sec. 5 shows that single fraction dominates the sequencing bill.

2.4 The Count Matrix and Estimand ✗

What comes out is a count matrix $\mathbf{Y} \in \mathbb{Z}_{\geq 0}^{N\times G}$ over N cells and G genes, plus an assignment $\mathbf{Z} \in \{0,1\}^{N\times L}$ of cells to the L *library elements* — the L distinct perturbations the library contains, one per guide RNA in a CRISPRi screen or one per barcoded deletion strain in a knockout collection. $Z_{i\ell} = 1$ means cell i was found to carry element ℓ, and recovering $\mathbf{Z}$ at all is the problem Sec. 3.5 is about; a cell whose element cannot be read is a row of zeros and is discarded. Because per-cell depth is low, the analysis does not treat a single cell as an observation of a perturbation's effect. Instead cells sharing a guide are pooled into a **pseudobulk** profile, and the estimand is

$$\theta_{\ell j} \; = \; \log_2 \frac{\mu_{\ell j}}{\mu_{\mathrm{ctrl},\, j}}, \qquad \mathbf{\Theta} \in \mathbb{R}^{L\times G}, \tag{1}$$

the log fold change of gene j under element ℓ against non-targeting controls. Two things follow, and they shape every later section. Single-cell resolution is used for *assignment and quality control*, not for the effect estimate — so the relevant precision question is how many cells share a guide, not how deep any one cell was read. And $\mathbf{\Theta}$ is dense: a genome-scale screen in one condition produces of order 6000×6000 coefficients, a directed map from every perturbation to every transcriptional consequence. That object, not a hit list, is what makes the screen worth running for model training.

3 Method Descriptions ✗

Two method families are candidates. Split-pool is instrument-free, its reagent cost does not scale with cells, and it extends to bacteria. Droplet is the established option, is available as a service at UIUC, and recovers roughly five times more mRNA per cell.

That depth gap sounds decisive and is not, because what sets precision is total UMIs per perturbation — cells × depth — not depth alone (section 4.2). Five-fold less depth is exactly offset by five-fold more cells, and split-pool buys cells far more cheaply than droplet buys depth. Two things break the symmetry. Below roughly 100 cells per perturbation the biological floor binds and no amount of depth substitutes (section 4.3), so the trade is only available above that floor. And depth is not free of cells either: at split-pool's 410 UMIs per cell the measurement is firmly shot-noise limited, so rRNA depletion — which raises usable depth without touching cell count — is worth more than either (section 3.4). The comparison that settles it is cost per *usable* cell, section 5.2, not depth.

3.1 Yeast-Optimized SPLiT-seq (Brettner et al., 2024) ✗

Brettner et al. [2] adapted Split-Pool Ligation-based Transcriptome sequencing to yeast, profiling 43,388 cells of *S. cerevisiae* (haploid and diploid) and *C. albicans* in total. That total spans several experiments and is not the number to design against; the per-run figures are. One barcoding run loaded 48 wells at ∼5,000 cells each, so ∼240,000 cells entered the chemistry; those were split into ten sublibraries, and the two that were sequenced returned ∼5,500 and ∼10,000 cells passing filter. Read that way the comparison with droplet is direct: one split-pool run barcodes an order of magnitude more cells than one Chromium channel's ∼20,000, and the sublibrary — not the run — is the unit that gets sequenced and priced. It is the reference protocol for the split-pool arm, so it is worth walking element by element: each step exists to solve a specific failure, and the reasons are what transfer to a new organism.

1. Fix before you digest. Cells are fixed in 4% formaldehyde for ∼18 h at 4 °C, then quenched into 100 mM Tris with RNase inhibitor. Formaldehyde crosslinks protein and nucleic acid, which does two jobs at once: it freezes the transcriptome, and it makes the cell rigid enough to survive as a reaction vessel through the later handling. The ordering is not arbitrary. Cell-wall digestion itself triggers the cell-wall-integrity and HOG stress pathways and induces stress-responsive genes within minutes [5], so a protocol that digested first would measure artifacts of its own sample preparation. Fixing first is what isolates the perturbation's signal from the handling.

2. Open the wall, then the membrane, separately. Zymolyase at 0.005 U/µL in 1 M sorbitol with 0.1 M Na_2EDTA, 37 °C for 15 min, digests the β-glucan layer; the sorbitol is osmotic support, since a wall-less spheroplast bursts in a normal buffer. Then 0.4% Triton X-100 on ice for 3 min permeabilises the membrane. The two steps are tuned against opposite failures: too little and reagents cannot get in, too much and the cell stops retaining its own cDNA and the whole single-cell premise collapses. Brettner et al. arrived at these values by screening 16 fixation/permeabilization combinations in parallel. That is itself an argument for the method: because a well's barcode is just another round-1 barcode, sixteen conditions are sixteen wells of one run, not sixteen runs. A droplet method has no equivalent — cell and reagent meet inside a droplet, so each condition needs its own channel, and the same screen is sixteen channels of reagent and sixteen library preps.

3. Round 1 — barcoded reverse transcription. Cells are counted, diluted, and distributed into 48 wells of a 96-well plate at ∼5,000 cells per well. Neither number is a tuned optimum. The 5,000 falls out of the recipe: a 25 µL reaction at the stated 200,000 cells/mL. The 48 is half the plate, and the paper gives no reason — but it has a consequence worth carrying, because round 1 then contributes 48 addresses rather than 96 and the realized barcode space for that run is $48 \times 96 \times 96 = 442{,}368$, half the 96^3 the protocol can deliver. Collision rates in section 5.4 are computed at the full 96^3; a run that loads half a plate doubles them.

Each well contains a well-specific barcoded primer pair: a barcoded 15-dT primer and a barcoded random hexamer, both at 3 µM, with Maxima H Minus reverse transcriptase and 7.5% PEG-6000. Both primers carry the same well barcode, BC_1 — neither is a **unique molecular identifier**. The random hexamer is six random bases used to *prime* anywhere on a transcript, not to label a molecule; the UMI arrives three rounds later, on the round-3 oligo, which is the only point at which molecules within a cell become distinguishable.

PEG-6000 is a molecular crowder. It occupies volume that enzyme and template cannot, so their effective local concentration rises and the bimolecular steps — reverse transcription here, ligation later — run faster than the absolute concentrations would suggest. The trade is viscosity, and mixing does get worse; the protocol inherits the 7.5% figure from mammalian SPLiT-seq and does not justify it, so it is a candidate for titration rather than a settled value.

The dual priming is a deliberate trade, and it is a *lever* because it is a ratio rather than a fixed requirement. Oligo-dT captures polyadenylated mRNA specifically; random hexamers prime on anything, including rRNA, and buy better 5′ coverage in exchange. Brettner et al. measure the consequence — recovered transcripts are 93.7% rRNA and only 5.75% mRNA — and say plainly what to do about it: "Users who prefer to study mRNA should lower the percentage of rRNA recovered by decreasing the ratio of random hexamer". The hexamer is not optional in general, since it is what carries the chemistry to transcripts with no poly(A) tail and therefore to bacteria; but for a 3′-counting screen in yeast the 5′ coverage it buys is unused, which makes this ratio the first thing to retune (Sec. 6).

4. Rounds 2 and 3 — split-pool ligation. Cells are pooled, filtered through a 15 µm strainer, and redistributed into a fresh 96-well plate. The filtration removes clumps, not contaminants: two cells stuck together take one path through the plates, receive one barcode and are read as a single cell, which is the split-pool equivalent of a droplet doublet. Straining is the only control on it, since there is no partition to over-fill. Each well holds a barcode oligo pre-annealed to a **linker strand**, a splint that positions the barcode against the 5′ end of the existing cDNA so T4 DNA ligase can seal the nick. After 30 min at 37 °C a **blocking strand** is added, which outcompetes the linker and retires that well's barcode so it cannot participate in any later round — without it, leftover oligos would scramble the barcode sequence. Pool, split, repeat for round 3, then terminate with EDTA. Three rounds of 96 give $96^3 = 884{,}736$ distinct paths available to the protocol; the round-3 oligo also carries the 10 nt UMI and a 5′ biotin.

5. Sublibraries, lysis, and a selection step. Cells are pooled and split into sublibraries of 5,000–20,000, which can be frozen.

What a sublibrary is for. Two independent reasons, and the first is the one that is easy to miss. *It is a fourth barcode.* Three rounds of 96 give 884,736 addresses, and a run puts ∼240,000 cells through them — enough that by the birthday argument of section 5.4 a substantial fraction collide. Splitting into ten sublibraries does not change any cell's BC1–BC2–BC3 path, but it does mean two cells sharing a path are still told apart if they were sequenced in different sublibraries, because the Illumina index distinguishes them. Ten sublibraries therefore multiply the effective address space by ten, and table 16 prices that trade directly against adding a fourth ligation round. *And it is the unit of sequencing.* A sublibrary is what gets lysed, prepped, indexed and loaded, so it is the granularity at which cells are converted into reads and into money. Brettner et al. sequenced two of their ten and froze the rest — which is how you match cells sequenced to the budget and the flow cell, and how a failed prep costs one tenth of a run instead of all of it.

The two reasons pull in opposite directions, which is why the size is a range rather than a number: smaller sublibraries mean fewer collisions but more library preps at $55 each (section 5.4).

Lysis is SDS plus proteinase K at 55 °C for 2 h. The biotin now earns its place: streptavidin beads capture only molecules that received the round-3 oligo, so anything that fell out of the barcoding scheme is washed away rather than sequenced.

How the second handle is added. At this point the molecule has a PCR handle at one end only — the one delivered on the round-3 oligo. The other end is the far end of the cDNA and carries nothing, so the molecule cannot yet be amplified. A **template-switching oligo** supplies the missing handle: when reverse transcriptase runs off the end of its template it adds two or three non-templated C residues, the TSO ends in rGrGrG and base-pairs with them, and the enzyme then copies the TSO. The handle is therefore *written by the polymerase*, not ligated on — which is what distinguishes it from every earlier step in this protocol, where a splint held two ends together for a ligase. With both handles present the library is amplified, fragmented, adapter-ligated and index-PCRed; the i7 index is added at that last step and is a separate sequence from either PCR handle, read on its own index read. That final index is barcode round 4 in all but name (Sec. 5.4).

What it delivers. 100–600 genes and 120–700 mRNA UMIs per cell; 25–50% of input cells recovered; barcode collisions well under 5%. Reagent cost is about \$0.005 per cell, but 94% of reads are rRNA and ∼75% of sequenced cells are discarded — so the cost that matters is 20× higher (Sec. 5).

Reading order follows assembly order. Each round ligates its barcode onto the free 5′ end of the growing strand, so later barcodes end up further from the cDNA. Read 2 therefore meets them newest-first: UMI, BC3, BC2, BC1, then cDNA. The reverse order is the intuitive guess and it is impossible — BC1 arrives on the round-1 RT primer, so it cannot sit further out than a barcode added afterwards. Gaisser et al.'s STARsolo offsets [7] pin the layout to the base (table 15).

3.2 Droplet Capture: 10x Chromium with In-Droplet Spheroplasting ✗

Jariani et al. [8] made the 10x Chromium platform work for yeast with a one-line change, and the reasoning behind it is more interesting than the change itself.

The obstacle is that the standard 10x workflow lyses cells inside the droplet on a thermal schedule tuned for a membrane, and a yeast cell wall does not cooperate. The obvious fix — spheroplast the cells first, then load them — is what Jackson and colleagues did, but it means feeding fragile wall-less cells through a microfluidic chip, where premature lysis releases mRNA into the shared buffer and contaminates every droplet downstream. Jariani et al. instead put zymolyase *into the reverse-transcription master mix*, replacing 1 µL of water with 100× enzyme, so cells enter the chip walled and intact and are digested only after they are already isolated. They verified the timing directly: cell counts hold on ice and through the 6 min of droplet generation, then drop 200-fold within 20 min at 53 °C. Wall digestion, lysis, and reverse transcription all happen in the same droplet, in that order, without the cell ever being vulnerable outside one.

The result was 6,118 cells across five conditions with median transcripts per cell of 528–1,261 — roughly triple Brettner et al.'s per-cell recovery, because oligo-dT-only capture on a bead is far more mRNA-selective than dT-plus-hexamer in situ. Library preparation cost fell to \$0.30 per cell against \$4.15 for the plate-based methods it replaced. Sequencing saturation was only 70–89%, meaning deeper sequencing would still have recovered more.

The chemistry has moved since. That work used 3′ v2. Version 3 roughly doubles UMI yield on 10x's own matched datasets, and the current generation at the UIUC core is GEM-X v3.1 on a Chromium X, quoted at ∼8% doublets at 20,000 cells recovered with CRISPR guide capture available as a catalog add-on. Any comparison drawn from the 2020 paper therefore understates droplet performance today, and Sec. 5 uses the newer figures.

Table 3: Provenance for every number drawn by hand in Figs. 1–3. The matplotlib figures are omitted because their generating scripts read the data directly. A **note** marks a number that needed a judgement call or that was corrected in review; an entry with no citation key is not backed by the library mirror and should not be quoted. Generated from `experiments/019-perturb-seq-costing/scripts/figure_sources.py`.

Element	As drawn	Source
scrnaseq-method-taxonomy.drawio		
Microfluidic chip (C1) throughput	`up to 96 per chip`	"It uses an integrated fluidic circuit to isolate and process up to 96 cells at once." [5] *Note:* Was drawn as ~160 cells, which is Gasch et al.'s 163-cell study total across four chip runs – a study total on a per-experiment axis. The capacity of one chip is the comparable quantity.
FACS into plates throughput	`~285 cells`	"we applied yscRNA-seq to 285 individual yeast cells" [9]

Table 3, continued

Element	As drawn	Source
Microdissection throughput	`~2,000 cells`	"Microdissection coupled with imaging has been used to profile the transcriptomes of Schizosaccharomyces pombe across a variety of environmental stressors and S. cerevisiae during aging (Saint et al., 2019; Wang et al., 2022)" ▲*Note:* NOT SOURCED TO A NUMBER. The review names the studies but states no cell count, and neither primary is in the mirror. The ~2,000 is an order-of-magnitude placeholder and is the one number in this figure that should not be quoted.
Droplet throughput range	`6,000-100,000`	"applied it to over 100,000 single cells from three crosses [upper bound; lower bound is Jariani et al.'s 6,118 cells]" [10]
Microwell array throughput	`up to 1,061,865`	"we used a microwell-based platform for single-cell isolation ... We profiled a total of 1.061.865 cells" [4] *Note:* Was ~75,000, taken from mDrop-seq – which is a droplet method, not a microwell one. Replaced with the microwell study that exists.
Combinatorial indexing throughput	`25,000-240,000 per run`	"Reverse transcription (in situ) was performed in 25 uL reactions in 48 wells of a 100 uL 96-well plate ... 200,000 cells/mL [= ~5,000 cells x 48 wells = ~240,000 barcoded per run; lower bound is Kuchina et al.'s 25,214-cell combined dataset]" [2] *Note:* The upper bound was 1,000,000, which is Gaisser et al.'s protocol capacity claim ("enables ... up to 1 million"), not a profiled count. Same correction as Fig. 4.
splitseq-barcoding.drawio		
round-1 wells and cells per well	`48 wells, ~5,000 cells each`	"Reverse transcription (in situ) was performed in 25 uL reactions in 48 wells of a 100 uL 96-well plate with each well containing wellspecific, barcoded primers ... 200,000 cells/mL" [2]
barcode space over three rounds	`963 = 884,736`	"Cells are then pooled and randomly split into a new 96-well plate, and a well-specific barcode is attached" [2] *Note:* Protocol capacity. The published run loaded 48 wells in round 1, so its realized space was 48x96x96 = 442,368 (Sec. 3.1).
sublibrary size	`5,000-20,000 cells`	"In experiment 3, the barcoded cells were evenly divided into 10 sublibraries. The two sequenced sublibraries returned ~5,500 and 10,000 barcoded cells that passed computational filtering." [2]
rRNA fraction of recovered transcripts	`93.7%`	"On average, the transcript proportions we recover are 93.7% rRNA, 0.005% tRNA, 0.04% ncRNA, and 5.75% mRNA." [2] `(md:70)`
droplet-10x-barcoding.drawio		
doublet rate at 20,000 cells	`~8% at 20,000 recovered`	"[10x Chromium specification for the GEM-X/v3.1 chemistry priced in Sec. 5; the protocol paper used 3' v2]" [8] *Note:* MANUFACTURER SPECIFICATION, not a measurement from a mirrored paper. Verify against the current 10x user guide before quoting.
digestion timing and cell-count drop	`6 min hold, 200-fold drop within 20 min at 53 C`	"cell counts hold on ice and through the 6 min of droplet generation, then drop 200-fold within 20 min at 53 C" [8]
bead barcode and UMI lengths	`16 nt CB, 12 nt UMI, 118 cycles`	"[10x 3' v3/GEM-X read layout; cycle count derived in experiments/019-perturb-seq-costing/scripts/read_structure.py]" [8] *Note:* Layout is the current GEM-X/v3.1 chemistry, not the v2 Jariani ran.

3.3 Compressed Perturb-seq (Yao et al., 2024) ✗

Yao et al. [11] is the direct precedent for deliberately putting *many* perturbations in one cell and recovering individual effects computationally, rather than treating multi-guide cells as a defect to be filtered out. It inverts the assumption underneath Sec. 4.5, so it is worth stating precisely what is compressed and what is assumed.

What is compressed. Not the library — the *samples*. Their framing: "if the perturbation effects are sparse . . . then we can measure a small number of random composite samples (comprising 'linear combinations' of individual sample phenotypes) and decompress those measurements to infer the effects of individual perturbations." Two routes generate composites: **cell-pooling**, deliberately overloading droplets with several singly-perturbed cells, and **guide-pooling**, infecting at high MOI so each cell carries a random set of guides. Guide-pooling is the one relevant to us. They ran it at MOI 10 in human THP1 cells over 598 target genes, achieving 2.50 guides per cell, and analyzed the 8,448 cells carrying three or more.

The decoding. **FR-Perturb** factorises the expression matrix by sparse PCA, runs LASSO on the resulting matrix of perturbations by factors, then multiplies back out to per-gene effects. The sparsity that makes this work is structural rather than incidental: regulatory effects are modular, so the effect matrix has low rank r and few non-zeros q per module, and the required number of composite samples is $O((q + r)\log n)$ rather than $O(n)$.

The assumption, exactly. Guide-pooling "requires the non-trivial assumption that the effect sizes of guides tend to combine additively in log expression space when multiple guides are present in the same cell." Interactions are not modeled in the first-order fit; they are argued to *cancel*, because each cell gets a random combination, so individual interaction effects average out "so long as there are an equal number of positive and negative interactions" — a condition they justify by citing genome-wide second-order fitness interactions in yeast. That is our own data domain, which makes the assumption more checkable for us than for them.

What it delivers, and what it does not. A stated 10-fold cost reduction for guide-pooling, with performance *increasing* in guides per cell and the best results in cells carrying four or more — they found no ceiling within the range

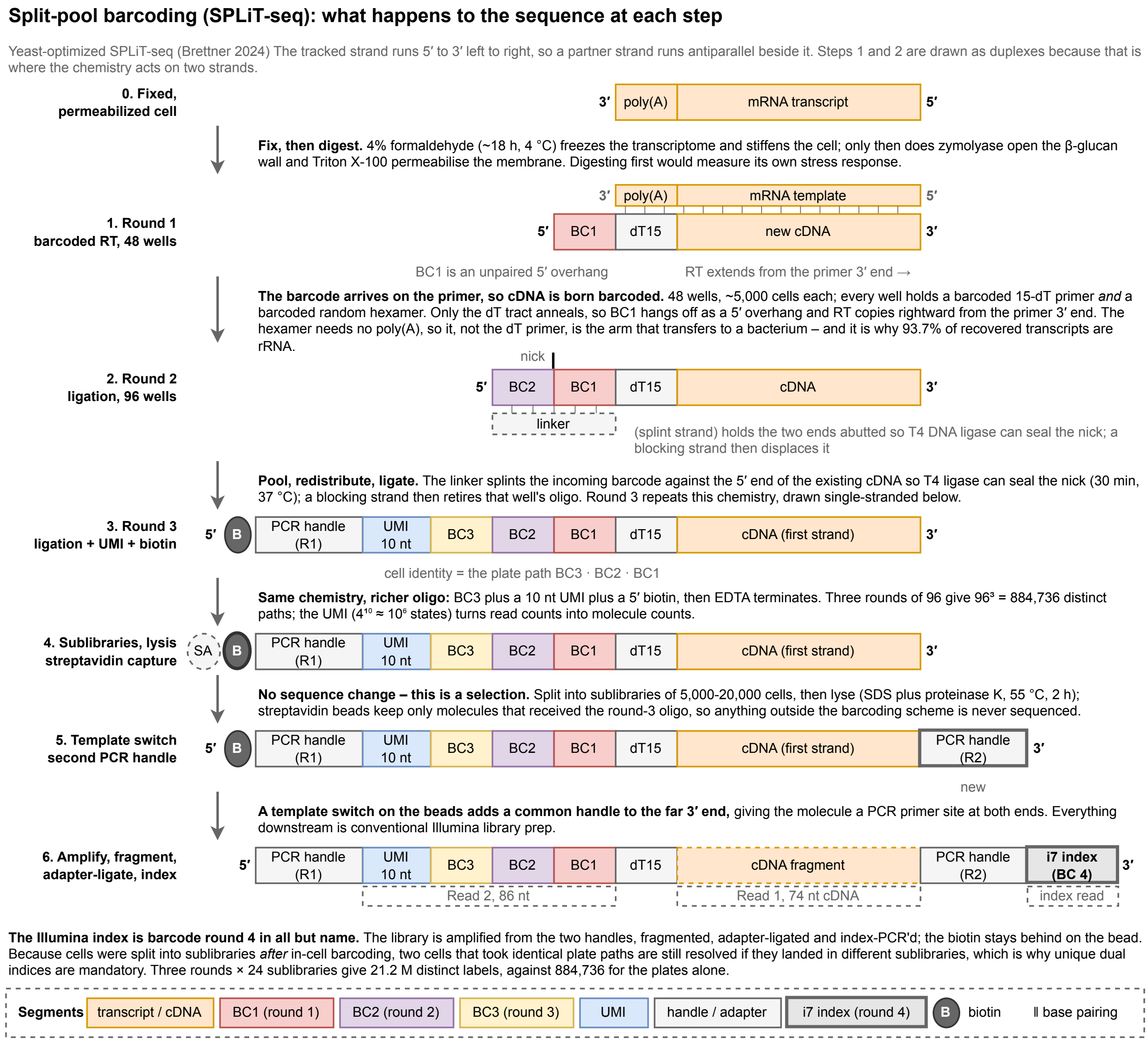

Figure 2: Cell identity in split-pool barcoding is assembled onto the transcript itself, which is what removes the need to ever physically isolate a cell. Seven strand states in the yeast-optimized SPLiT-seq workflow of section 3.1 [2], each arrow annotated with the chemistry that produces the next. Strands are drawn 5′ left to 3′ right, so the barcode block grows leftward with each round while the cDNA body stays fixed — the visual point being that a cell's address is written in three installments, and a cell is identified by the *path* it took through three 96-well plates rather than by ever having been alone in a well. That is the whole economic argument for the method: 96^3 distinct paths cost three plates, not 10^6 containers. Two states are drawn double-stranded because that is where the mechanism is easy to get wrong. At round 1 the mRNA template lies antiparallel above the tracked strand, showing BC1 as an unpaired 5′ overhang, and showing that the random hexamer rather than the poly-dT primer is the arm that carries the method to transcripts without a poly(A) tail — the single change that lets the same chemistry work in bacteria [6, 7]. At round 2 the linker strand base-pairs underneath and a tick marks the nick T4 ligase seals; ligation is templated by that splint, which is why an unretired barcode from a previous well would scramble the address and why a blocking strand follows every round. The 5′ biotin added in round 3 is not a label but a selection handle: streptavidin capture at the next step discards every molecule that fell out of the scheme, so incompletely barcoded material is removed before it can consume sequencing. Composed from the protocol as described in section 3.1, not traced from any published figure; every parameter shown is sourced there. Drawn at true Nature print size in `notes/assets/drawio/splitseq-barcoding.drawio`.

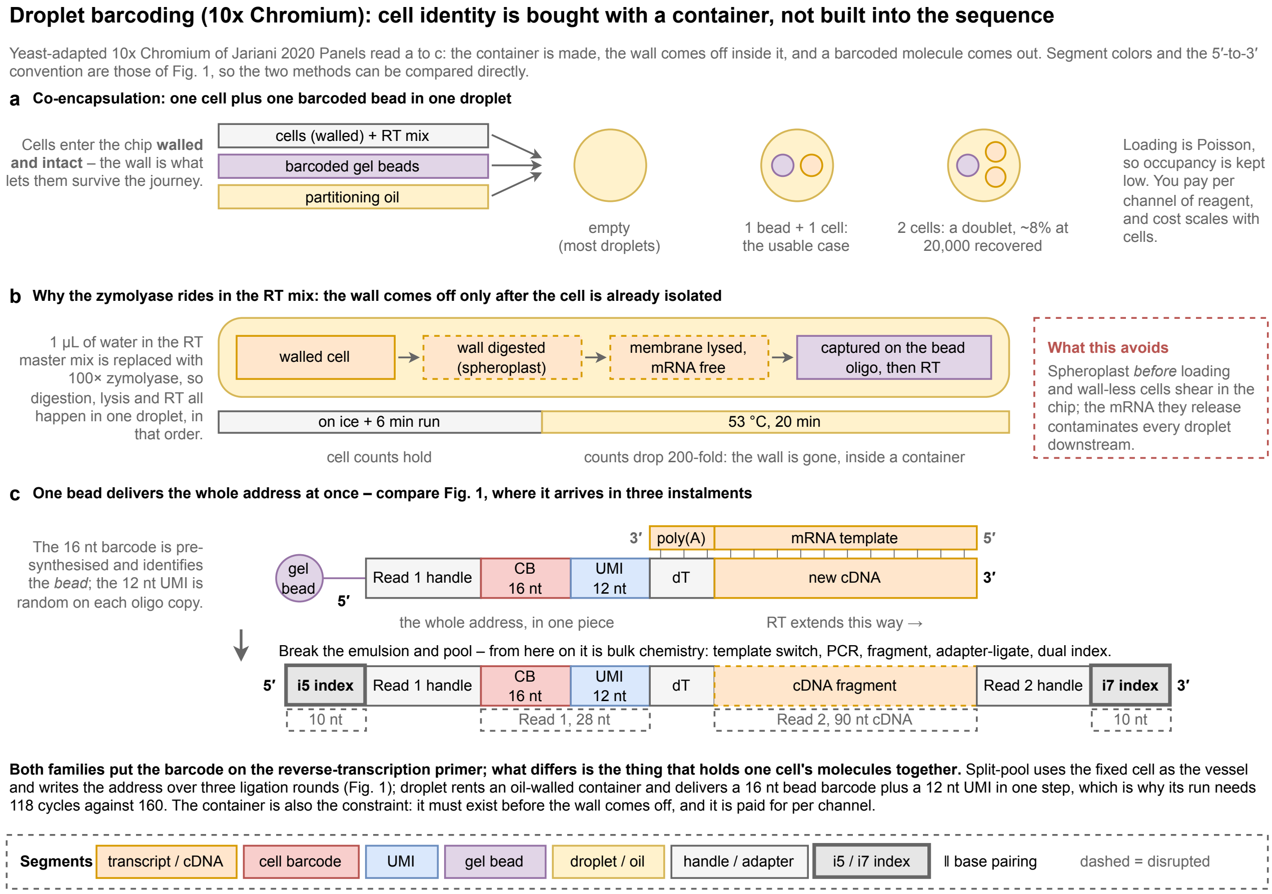

Figure 3: Droplet methods buy cell identity with a physical container, which is what lets the whole barcode arrive at once — and what turns the yeast cell wall into a scheduling problem rather than a chemistry one. The yeast-adapted 10x Chromium workflow of section 3.2 [8]. **(a)** Cells, beads and oil meet at a junction and partition into droplets. Identity here is positional rather than sequential: every molecule tagged inside a droplet carries that droplet's bead barcode because the oil wall keeps it that way. Loading follows Poisson statistics, so occupancy is kept deliberately low — most droplets are empty, and the doublets that survive that choice are two cells reported as one. The container is bought per channel of reagent and its cost scales linearly with cells, which is precisely the economics fig. 2 avoids by never isolating anything. **(b)** Jariani et al.'s change is one line of the protocol and is entirely about *ordering*: 1 µL of water in the reverse-transcription master mix is replaced with 100× zymolyase, so cells enter the chip walled and are digested only once they are already alone. The timing was verified rather than assumed — counts hold on ice and through the 6 min of droplet generation, then fall 200-fold within 20 min at 53 °C, by which point every cell is sealed in its own container. The alternative of spheroplasting before loading feeds wall-less cells through a microfluidic chip, and mRNA leaking from the ones that shear contaminates every droplet downstream, which is a whole-run failure rather than a per-cell one. **(c)** The resulting molecule, drawn in the notation of fig. 2 so the two can be read against each other. The bead oligo *is* the reverse-transcription primer, so as in split-pool the barcode arrives on the primer and only the poly-dT tract anneals — but the entire address, a 16 nt bead barcode plus a 12 nt UMI, is delivered in one piece instead of over three ligation rounds. Once the emulsion is broken the chemistry is bulk, and the read layout that results needs 118 cycles against split-pool's 160 (`experiments/019-perturb-seq-costing/scripts/read_structure.py`). That layout is the current GEM-X/v3.1 chemistry priced in section 5; Jariani et al. demonstrated on 3′ v2, whose Read 1 is 26 nt. Composed from the protocol as described in section 3.2, not traced from any published figure. Every number drawn by hand in this figure is sourced in table 3.

tested and expect further gains from "even higher efficiency with a greater degree of overloading." The honest limits are equally clear. Pairwise interactions were not recovered at all: "none of these effects was significant in either screen due to insufficient power ... even with a lax significance threshold ($q < 0.5$)." And higher-order additivity is essentially untested — "very little is known about the structure of higher-order interaction effects and how they systematically combine within a cell."

> **Why this matters here.** It converts multiplexing from a defect into a design variable, and supplies the constant that Sec. 4.5 needs: ∼400 cells per pairwise interaction, four times the 100-cell first-order floor. But it recovers *main effects* efficiently by assuming interactions cancel — and interactions are what metabolic engineering needs. The two goals pull in opposite directions, which is the tension Sec. 4.5 resolves.

3.4 Ribosomal RNA Depletion ✗

Roughly 85% of a yeast cell's RNA is ribosomal, and Brettner et al. recover 93.7% rRNA against 5.75% mRNA. Ninety-four of every hundred reads therefore buy ribosomal sequence. Since sequencing is 81% of the split-pool budget (Sec. 5.5), this single fraction is the largest cost lever in the document — worth about $99,000 at genome scale — and it is the reason depletion is treated here as a method component rather than an optimization.

Where it happens in the workflow. This is worth stating precisely, because there are three plausible points and only one is right. Depletion is *not* performed on the cell before capture, and *not* after Illumina indexing. It acts on the **amplified cDNA library**, after reverse transcription and the in-cell barcoding rounds, and before fragmentation and index PCR:

capture → barcode → lyse → cDNA → **deplete** → fragment → index → sequence

Two constraints force that position. Ribosomal RNA cannot be removed before capture without also stripping the ribosomes whose footprint the in-situ chemistry depends on, and the cell barcode must already be attached or depletion would discard cell identity along with the rRNA. Equally it must precede indexing, because the entire point is to avoid paying to sequence those molecules. Brandner et al. [3] do exactly this: the purified cDNA library is incubated with recombinant Cas9 and a pool of guides targeting rRNA, cleaving rRNA-derived cDNA so it no longer amplifies. Wang et al.'s M3-seq [12] places the same operation at the same point and builds a whole platform around it, which is the strongest evidence that this position in the workflow is the right one rather than merely a convenient one.

Two independent levers, and they compose. The first is free. Brettner et al. prime with both oligo-dT and random hexamers, and it is the hexamers that recover rRNA; the 5′ coverage they buy is worthless for a 3′-counting screen, so shifting the hexamer:dT ratio costs nothing but a titration. The second is the Cas9 depletion above, which adds reagents but little labor.

How much it is worth, and the caveat. Brandner et al. took *E. coli* from 4.6% to 60.2% mRNA, a 13-fold improvement, and median mRNA transcripts per cell more than doubled from 54 to 115. But the same protocol in *P. putida* moved only 2.2% to 10.4%. Depletion efficiency clearly does not transfer between organisms for free, since the guide pool is designed against one species' rRNA and against the coverage non-uniformity of one protocol. Yeast should be better than either bacterium — it starts at ∼85% rRNA rather than >95%, and its rRNA genes are fewer and better annotated — but the achieved figure has to be measured, and it is the single most valuable number the pilot can return (Sec. 6).

> **Treat depletion as mandatory, not optional.** Without it the genome-scale screen costs $157 k and sequencing dominates; with it, $58 k and the budget becomes balanced. No other change in this document moves the total by as much.

3.5 Perturbation Identity Capture ✗

Both families above assume the guide can be read out of the cell. In yeast, as our library stands, it cannot.

Guide RNAs are transcribed from RNA polymerase III promoters — *SNR52* upstream, *SUP4* terminator — so they are neither capped nor polyadenylated and terminate on a T-tract. A 3′ poly(A) capture assay, which is what both Brettner et al.'s oligo-dT priming and 10x's bead oligo are, sees nothing. Our own MAGIC library uses exactly this architecture and reads out by plasmid amplicon sequencing, not RNA [13]; our reprocessing of it is `experiments/016-lian-magic-reprocess`.

The mammalian workarounds do not transfer. 10x direct capture requires a modified sgRNA scaffold carrying a capture sequence; CROP-seq [14] relies on the guide cassette being duplicated into a lentiviral 3′ LTR, which is a retroviral trick with no yeast equivalent.

The gap has closed for deletions, and that changes the risk. The 2024 review states plainly that no genome-scale system in yeast lets a genetic perturbation be traced during scRNA-seq [5]. That was true when written and is no longer, because the same group built one the following year [4]. The construction is worth following closely, because it is the design this section was going to propose.

They rebuilt the yeast knockout collection so its genotype tag is *transcribed.* In the classical YKOC a *TEF1* promoter drives KanMX and two 20 bp barcodes sit outside the transcript, which — in their words — "renders the genotype identity invisible to transcriptomic readouts." They replaced the KanMX marker with *URA3* and shortened the heterologous terminator from 262 nt to 43 nt, which brings the Downtag barcode inside the *URA3* 3′UTR, and added a 5 nt clone barcode after the stop codon. The tag is now a polyadenylated Pol II transcript, so an oligo-dT assay sees it. That is exactly the architecture proposed below, executed at genome scale: 4,162 barcoded genotypes, of which more than 3,500 were recovered across 1,061,865 profiled cells, 710,952 of them passing QC with a genotype assigned.

Two numbers from it matter more than the rest. Genotype identity was assigned for **more than 71% of cells** — against the 21–25% Brandner et al. reach in bacteria — which is the parameter q that governs everything in section 4.5, and it is three-fold better than the figure this document has been budgeting with. And it was not free: the assignment combines the transcriptome with a *separate* one-step PCR amplicon library off the *URA3* 3′UTR. Targeted enrichment is doing the work in both systems, so it should be costed as a second library, not as a primer.

What is still open for us. Their perturbation is a chromosomal deletion with the tag built into the deletion cassette. Ours is a plasmid-borne CRISPRi guide, so the tag has to ride on the plasmid, and the construct does not exist. The architecture transfers directly, and its bacterial counterpart is also precedented: Brandner et al. [3] built a 316 bp non-coding transcript carrying a 12 bp barcode at its 5′ end, a poly(A) tail and a dedicated enrichment primer site, using an inert gene body as filler, and raised correct assignment from 5.5% to 21–25% by adding the enrichment primer at cDNA amplification. A design choice worth taking: make the barcode *be* the guide's own 20 nt spacer, placed in the 3′UTR of a Pol II reporter, so the guide-to-barcode mapping is known by construction — that removes a linkage sequencing run and a whole class of bookkeeping error.

Still the critical path, but no longer the risky one. Every cost in Sec. 5 assumes guide identity is recoverable, and for a plasmid-borne CRISPRi guide it is still unbuilt. What changed is that the same thing has now been done in yeast at genome scale with a published construct and a 71% assignment rate, so this is a construction and validation task against a working template rather than an open research question. The number to reproduce in the pilot is q, because section 4.5 shows it enters the multiplex ceiling as q^k.

4 Theoretical Limits ✗

The question this section answers: given how much mRNA a yeast cell contains and how little of it any method recovers, how many cells must share a perturbation before its mean effect is measurable — and how does that change when a cell carries more than one perturbation?

Two sources of noise, and only one of them is fixable by sequencing. Every argument below turns on a distinction worth making once, plainly. **Shot noise** — also called counting or sampling noise — is the irreducible randomness of *counting.* Sequencing a library samples molecules at random, so a gene present at a true rate μ is observed as a count that fluctuates Poisson-like around μ, with variance equal to μ and therefore a relative error $1/\sqrt{\mu}$. It is a property of the measurement, not of the cell, and it is the one kind of noise that buying more reads removes. **Biological overdispersion** is the rest: two genetically identical cells in the same flask genuinely differ in how much of a gene they are expressing, because of cell-cycle phase, metabolic state and transcriptional bursting. Sequencing each cell to infinite depth measures each cell perfectly and leaves this variation completely untouched; only counting *more cells* averages it down. Keeping the two apart is what section 4.7 is for — the whole design reduces to which of them is currently binding.

4.1 Transcript Content per Cell ✗

Every capture-efficiency claim needs this denominator, and the literature does not agree on it.

The disagreement is real and is left visible rather than averaged away: the 3,000 genes at 3.5 molecules each implied by the review works out to ∼10,500 mRNA per cell, materially below the 20,000 floor of the range Jariani et al. cites

Table 5: mRNA content per cell. This is the denominator for every capture-efficiency claim in the document. [a] BNID 112795, `https://bionumbers.hms.harvard.edu/bionumber.aspx?id=112795` – the one row here that is *not* in the torchcell-library mirror. No mirrored bacterial paper states a total mRNA content, only what its assay detected, which is a different and ∼30× smaller quantity. Every other row is mirror-backed and separately citable. **The yeast rows disagree and are left that way.** Jariani et al.'s 20,000 floor, Jackson et al.'s 40,000–60,000 and the ∼10,500 implied by the review's own 3,000 genes × 3.5 molecules span six-fold. The document works at ∼30,000; because p_j enters eq. (5) only through the shot-noise term, and section 4.7 finds both candidate platforms firmly shot-noise limited, this choice does move the cells-per-perturbation numbers roughly in proportion.

Organism	Quantity	Value	Source
S. cerevisiae	Total mRNA molecules per cell	**20,000–60,000**	Jariani 2020 (`md:31`)
	Working figure (Brettner)	∼30,000	Brettner 2024 (`md:70`)
	Total mRNA molecules per cell	40,000–60,000	Jackson 2020 [15]
	Implied by 3,000 genes × 3.5 molecules	∼10,500	Nadal-Ribelles 2024 (`md:107`)
	Mean UMIs per detected gene, per cell	3.46	Nadal-Ribelles 2019 [9]
	Capture rate, 10x 3′ v2	3–5%	Jackson 2020 [15]
	Total RNA per cell (∼85% rRNA)	0.7–1 pg	Nadal-Ribelles 2024 (`md:49`)
	Protein-coding genes	∼6,000	SGD / R64
E. coli	Total mRNA per cell, **rich (LB)**	**∼7,800**	Bartholomäus 2016[a]
	Total mRNA per cell, minimal medium	∼2,400	Bartholomäus 2016[a]
	Total mRNA per cell, medium growth rate	∼3,000	Taniguchi 2010 [16]
	rRNA share of all transcripts	**>95%**	Brandner 2025 (`md:71`)
	Protein-coding genes	∼4,400	K-12 MG1655
Mammalian	Total mRNA molecules per cell	50,000–300,000	Jariani 2020 (`md:31`)

and well below the 40,000–60,000 Jackson et al. give.

This choice is not harmless, and the working value mixes two sources. It is tempting to treat the spread as a detail, and it is not. The design equation reaches this quantity through p_j, the transcriptome share of the target gene, and p_j enters only the shot-noise term — where both candidate platforms sit (section 4.7), so n^* scales with it almost directly. Taking 3.5 molecules per gene against each candidate denominator, at $d = 410$ and $\varphi = 1.1$, gives 137 cells per perturbation at 10,500 total mRNA, 359 at 30,000, and 699 at 60,000. That is a five-fold spread in the number the whole budget is built on, and it is larger than most of the effects this document argues about.

Worse, the working value is not internally consistent. The document uses $p_j = 3.5/30{,}000$, which pairs the review's per-gene figure with Brettner's total — but the 3.5 was derived *against* the review's own ∼10,500, so used self-consistently it would give $p_j = 3.3 \times 10^{-4}$ and cut every cells-per-perturbation figure by a factor of 2.9. The 30,000 pairing is the conservative one, which is why it is kept, but it is a choice and not a measurement, and table 10 accordingly marks p_j derived. Measuring the transcriptome share of a few representative target genes directly in our own pilot data is cheap and would retire the single largest unforced uncertainty in section 5.

The comparison that does matter is across organisms — a yeast cell holds roughly four times the mRNA of an *E. coli* cell in rich medium over ∼1.4× the gene count, which is why the same split-pool chemistry recovers ∼410 mRNA UMIs per cell in yeast and 235–397 in bacteria.

4.2 The Cells-Versus-Depth Frontier ✗

Plotting every published yeast and microbial run on one pair of axes — cells profiled against mRNA UMIs recovered per cell — shows that the field has not found a method that is good at both (fig. 4; per-study values in table 6).[1] Depth does fall as cell count rises, so there is a trade-off, but it is a much weaker one than the picture suggests at a glance. Cells profiled span 3.6 orders of magnitude across these ten studies and depth only 1.6, and moving from the smallest study to the largest multiplies cells by 3,726 while dividing depth by just 9.8. Both endpoints happen to be Nadal-Ribelles et al., six years apart [4, 9].

The consequence is worth stating plainly, because the figure's diagonals invite the opposite reading. *Total* UMIs is not conserved along this cloud: it spans 3.2 orders of magnitude, which is nearly all of the variation in cell count. Larger studies do not redistribute a fixed amount of signal over more cells; they collect more signal. The published record is therefore not a frontier of equal-value points, and no method here has been shown to be a better bargain than another by sitting further out along a diagonal.

What the diagonals do encode is a statistical fact about *design*, not a summary of what has been published. In the shot-noise limit the precision of a perturbation's estimated mean depends on how many molecules were counted in total,

[1] Plotted studies, in order of cells profiled: Nadal-Ribelles et al. 2019 [9], Urbonaite et al. [17], McNulty et al. [18], Jariani et al. [8], Kuchina et al. [6], Jackson et al. [15], Brettner et al. [2], Brandner et al. [3], Boocock et al. [10] and Nadal-Ribelles et al. 2025 [4]. Tabulated but not plotted, for want of a per-cell molecule count: Gasch et al. [19], Ma et al. [20] and Dohn et al.

not on how they were distributed over cells, so for a fixed sequencing spend, moving along a diagonal is free and should be decided on other grounds — instrument access, cost structure, whether a perturbation can be read out at all. That is the lever section 5 pulls. It is a claim about the arithmetic in eq. (5), and the studies plotted here neither support nor contradict it, because none of them was designed against a common budget.

Two features carry into the rest of this document, and the second has changed recently. Published runs span 9×10^5 to 1.3×10^9 total UMIs. At the low end that is three orders of magnitude inside what a single modern flow cell delivers (section 5.3), so those studies were bounded by protocol throughput and budget rather than by sequencing capacity — which is why platform choice, not sequencer choice, is what a small screen design turns on. That is no longer true at the top end. Nadal-Ribelles et al.'s 1.06-million-cell atlas needs 1.3×10^9 UMIs, within a factor of three of the 3.2 billion read pairs in one NovaSeq X 25B lane *before* counting duplication or the rRNA fraction, so at genome scale in yeast the sequencer has started to bind. section 5 prices exactly that regime.

Second, four methods now carry a perturbation readout, and they no longer sit at one end of anything: they run from Jackson et al.'s 38,225 cells to Nadal-Ribelles et al.'s 1.06 million, and from 325 UMIs per cell to 2,809. The largest genetic screen in this figure is a yeast one, and it used neither of the two chemistries this document has been comparing — it used a microwell array (section 3.5). Split-pool versus droplet is therefore not a throughput question at the scales anyone has run; it is a question of depth and of cost per *usable* cell, which is section 5.

Table 6: Published single-cell RNA-seq runs in yeast and other microbes. † values taken from the Nadal-Ribelles review's summary table because the primary paper is not in our mirror, so they are second-hand and should be verified before being cited. ‡ the quoted UMI range spans separate *experiments* of deliberately differing depth rather than conditions within one run. **No range in the UMI column is a dispersion statistic over cells**; all are min–max ranges of per-condition or per-experiment summaries. **A dash in the UMI column is deliberate**: Gasch et al. used a protocol that carries no UMI, and Ma et al. report only genes, so neither has a molecule count to state and none is invented for them – their measured quantity appears under Genes/cell instead. **Basis** states how the single plotted value was obtained: *reported* = stated by the source; *midpoint* = midpoint of a reported range, a declared convention where no per-condition value exists; *median-of* = median of the reported per-condition values; *weighted* = cell-weighted mean of reported per-dataset values; *derived* = computed from two quantities the source states separately, never printing the one needed here. "Pert." = has a perturbation readout.

Study	Platform	Isolation	Cells	mRNA UMIs/cell	Genes/cell	Pert.	Basis
Gasch 2017	Fluidigm C1	plate	163	–	735–5,437	–	rep.
Nadal-Ribelles 2019	FACS + STRT-seq	plate	285	11,760	3,399	–	deriv.
Jackson 2020	10x 3′ v2	droplet	38,225	2,250	695	yes	rep.
Jariani 2020	10x 3′ v2 (in-drop zym.)	droplet	6,118	528–1,261	–	–	med-of
Urbonaite 2021	yeastDrop-Seq	droplet	844	840–1,336	443–619	–	midpt.
Dohn 2021†	mDrop-seq	droplet	74,814	–	193–967	–	rep.
Boocock 2025‡	10x 3′ v2 + v3	droplet	100,220	1,514–6,559	–	yes	wtd.
Nadal-Ribelles 2025	Singleron GEXSCOPE HD (microwell)	microwell	1,061,865	1,200	550	yes	rep.
Brettner 2024‡	SPLiT-seq (yeast)	split-pool	43,388	120–700	100–600	–	midpt.
Kuchina 2021	microSPLiT	split-pool	25,214	397	230	–	rep.
McNulty 2023	ProBac-seq	droplet	2,784	325	241	–	rep.
Ma 2023	BacDrop	droplet	44,140	–	30–127	–	rep.
Brandner 2025	mapSPLiT	split-pool	76,068	325	161	yes	rep.

4.3 Cells per Perturbation ✗

Two requirements act independently, and the larger of the two binds.

Constraint 1 — a biological floor. Averaging out cell-cycle phase and metabolic state takes cells, and no amount of sequencing depth substitutes for them. The field standard is stated explicitly by Brandner et al. [3]: perturbations are retained only with at least 100 cells. Brettner et al.'s own heuristic is 500 cells per condition. The strongest corroboration is a design choice rather than a rule: the genome-scale yeast screen of Nadal-Ribelles et al. [4] landed at an average of 93 and 108 cells per genotype in its two conditions (medians 77 and 87). A million-cell budget spread over 3,500 mutants converges on the same hundred-cell scale that the convention asserts, arrived at independently.

Constraint 2 — pseudobulk depth. Pooling n cells at depth d gives $K \approx ndp_j$ UMIs of gene j. For counts, $\mathrm{Var}(\log K) \approx 1/K$, so resolving a $\log_2$ fold change Δ at power $1-\beta$ and level α requires

$$K \geq \left(\frac{z}{\Delta \ln 2}\right)^2, \qquad z = z_{1-\alpha/2} + z_{1-\beta} = 2.80, \tag{2}$$

which is ∼16 UMIs of that gene for a two-fold change, ∼65 for 1.4-fold, ∼159 for 1.25-fold. This is a floor, not an estimate: it counts only sampling noise and ignores biological variance between cells, which is exactly why the 100-cell floor is a separate and independent requirement.

Two things follow. The 488 cells that the depth constraint demands of Brettner et al.'s chemistry reproduces his own 500-cell heuristic from an independent derivation, which is reassurance that the framework is calibrated. And depth

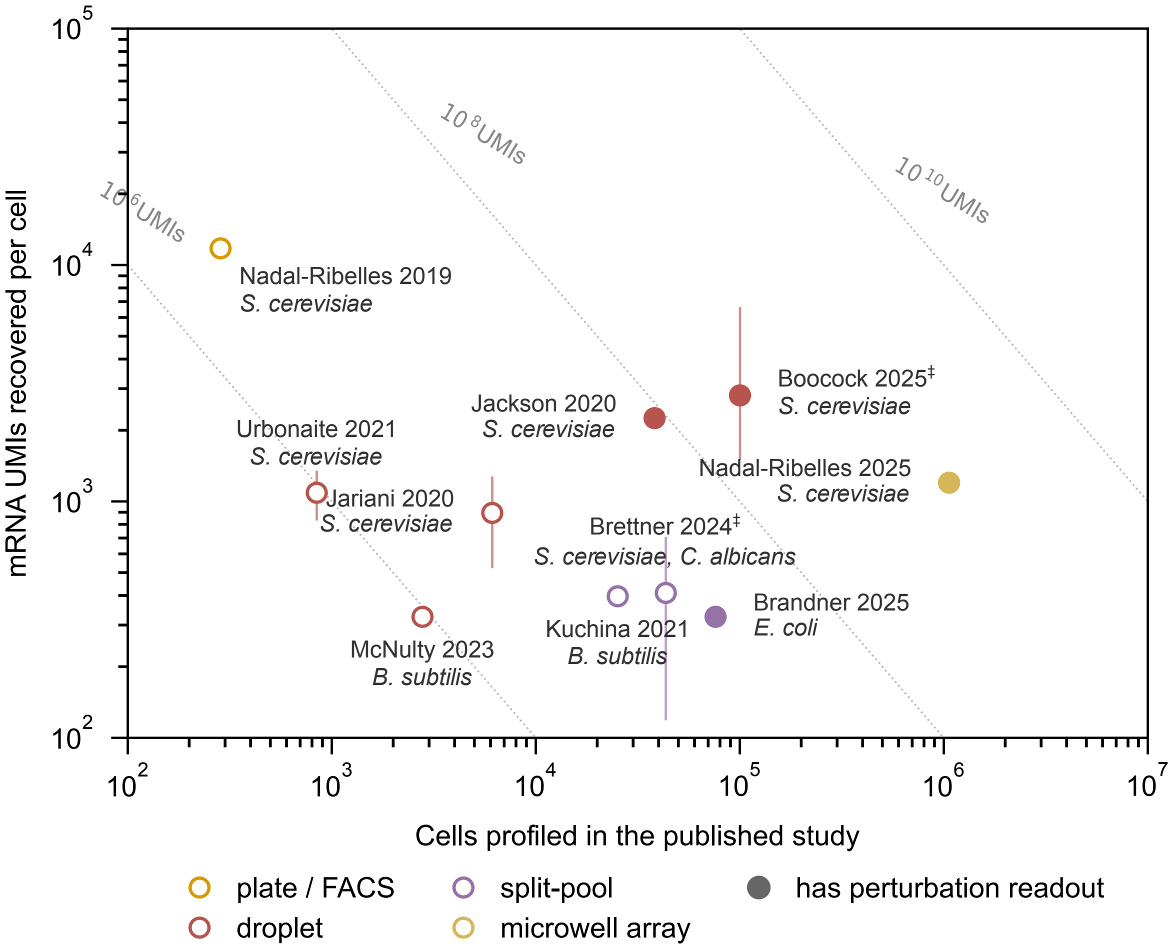

Figure 4: Published microbial single-cell RNA-seq trades cells against depth, but far more weakly than the diagonals suggest. Cells profiled in each published study (x) against mRNA UMIs recovered per cell (y), organism in italic beneath each label; marker color is the isolation principle and filled markers denote a method with a perturbation readout. Across these ten studies cells span 3.6 orders of magnitude and depth only 1.6, and the largest study holds 3,726 times the cells of the smallest at just 9.8 times less depth — so *total* UMIs is not conserved along the cloud, but spans 3.2 orders of magnitude, nearly all of the variation in cell count. Bigger studies collect more signal rather than spreading a fixed amount of it thinner. **The dotted diagonals are therefore a design tool, not a summary of this data.** They are iso-lines of constant total UMIs, and that product is what sets the shot-noise floor on a perturbation's estimated mean, so at a *fixed budget* sliding along one is free and only moving up-and-right buys precision. None of these studies was designed against a common budget, so none of them tests that. **The vertical bars are not error bars.** No plotted study reports a dispersion statistic over cells. Each bar is the min–max range of the *per-condition summary values* a study reports — medians or means across its own treatment conditions — so it describes how far the assay moved between conditions, not how much cells vary within one; ‡ marks a bar that spans separate experiments of deliberately differing depth, which is weaker still. Studies reporting a single value have no bar. **Two methods are absent from the depth axis by design.** Gasch et al. used Fluidigm C1 with SMART-Seq, which carries no UMI, and Ma et al. report only genes per cell; neither has a molecule count, so neither is given one. Their measured quantities appear under Genes/cell in table 6. This matters because the quantity most easily confused with depth here is the gene count, and reading one as the other moves a method bodily along the frontier. Published runs span 9×10^5 to 1.3×10^9 total UMIs. The smaller ones sit three orders of magnitude short of the 3.2 billion read pairs a single NovaSeq X 25B lane delivers (fig. 7) and were limited by protocol throughput and budget rather than by the instrument; the largest is within a factor of three of one lane before duplication and rRNA are counted, so at that scale the sequencer does bind. Plate methods buy depth by giving up cell number, bottoming out at a few hundred cells; droplet methods run deeper than split-pool at comparable cell counts, the largest being the ~100,000-cell eQTL run of Boocock et al. [10]; split-pool (Brettner et al., microSPLiT, mapSPLiT) sits at 10^4–10^5 cells and ~300–400 UMIs, on the order of 1% of the 20,000–60,000 mRNA molecules a yeast cell carries; and the microwell array of Nadal-Ribelles et al. [4] reaches 1.06 million cells at 1,200 UMIs. That last point is a genome-scale yeast Perturb-seq — 3,500 barcoded deletion mutants read out in one atlas — and it is the largest genetic screen here (section 3.5). ‡ Boocock et al.'s seven datasets were sequenced to deliberately different depths, so the bar spans the per-dataset medians (1,514–6,559) and the marker is placed at their cell-weighted mean, 2,809 UMIs, from Supplementary file 1 Table S4 — weighting by cells is what keeps the point describing the same 100,220 cells its x-coordinate claims. Nadal-Ribelles et al.'s depth is the one derived value (3,399 genes per cell × 3.46 UMIs per detected gene = 11,760); the source states both factors but never their product. Generated by `experiments/019-perturb-seq-costing/scripts/plot_method_landscape.py`.

Table 7: Cells per perturbation at the standard pseudobulk tier (2×10^5 mRNA UMIs). The 100-cell biological floor and the depth requirement are independent; the larger binds.

Method	mRNA UMIs/cell	Cells/pert.	Binding constraint	Cells for 6,000 genes
SPLiT-seq as published	410	**488**	sequencing depth	2,928,000
SPLiT-seq + rRNA depletion	861	**233**	sequencing depth	1,398,000
10x v2 (Jariani)	894	**224**	sequencing depth	1,344,000
10x v3 (Jackson)	2,000	**100**	biological floor	600,000

buys you out of cells only until the 100-cell floor binds — past 10x v3 chemistry, extra depth is wasted on this question and should be spent on more perturbations instead.

4.4 Effect Size of a Single Knockdown ✗

Equation 2 takes the fold change Δ as given, but Δ is an empirical property of yeast regulatory biology, and it is the parameter the whole design is most sensitive to. It is measured here rather than assumed, from two datasets already in the library mirror: Kemmeren et al.'s single-deletion expression compendium [21] and Sameith et al.'s double-deletion compendium [22], both built locally and analyzed below.

Table 8: Measured response of the transcriptome to a gene deletion. A single deletion moves a few hundred genes by 1.25× but only ∼10–15 by 2×, so designing at two-fold targets a few percent of the response. The median responding gene moves ∼1.34×, and because the power coefficient A scales as $|\Delta|^{-2}$ this multiplies every cell requirement by ∼5.6 relative to the two-fold assumption. Doubling the perturbation count roughly doubles the responder count at 1.25× and trebles it at 2×, without shifting the median response — more perturbations make a cell noisier, not its individual effects larger. Generated from `experiments/019-perturb-seq-costing/scripts/effect_size_analysis.py`.

Compendium	Strains	Median responding genes at			Median $\lvert\Delta\rvert$, responders	
		1.25×	1.5×	2×	$\log_2$	fold
Kemmeren 2014, single deletion	1,484	269	55	15	**0.42**	1.34×
Sameith 2015, single deletion	82	220	42	10	**0.42**	1.34×
Sameith 2015, **double** deletion	72	460	112	30	**0.44**	1.36×

Three findings, and the first two change the arithmetic of the whole document.

A deletion moves hundreds of genes, but almost none of them by two-fold. The median single deletion moves 220–269 genes by more than 1.25×, but only 10–15 by more than 2× (table 8). Designing a screen at a two-fold threshold therefore targets a few percent of the transcriptional response and discards the rest. Kemmeren's 1,484 single deletions and Sameith's 82 agree closely, which is reassurance that this is a property of yeast rather than of one platform.

The median responding gene moves 1.34-fold, not 2-fold. Among genes that do respond, the median $|\Delta|$ is 0.42 in $\log_2$ units, and the three compendia agree to within 0.02 despite differing in size by a factor of eighteen. This is the number Eq. 2 wants. Because the power coefficient scales as $|\Delta|^{-2}$, replacing the nominal two-fold with the measured value multiplies every cells-per-perturbation figure in this document by ∼5.6. It is the single largest correction in the analysis, and it moves in the expensive direction.

More perturbations make a cell noisier, not its effects larger. Sameith's doubles move roughly twice as many genes as its singles at 1.25× (460 against 220) and three times as many at 2× (30 against 10), while the median response among responders is unchanged (0.44 against 0.42). A second deletion recruits additional genes into the response rather than amplifying the existing ones.

One caveat on which way this is likely to be biased. Sameith et al. did not sample gene pairs at random: the double-deletion panel is built from genes with regulatory roles, chosen because they were expected to interact. Its responses should therefore sit *above* what a random pair of genes would produce, and its singles above a random single. Kemmeren's 1,484 deletions are the closer thing to an unbiased draw, and its median response, 0.42, is essentially identical to Sameith's 0.42 — so the bias does not appear to move the median *effect size* much. Where it should be assumed to matter is the responder *counts*, and the doubles in particular: 460 responding genes per double deletion is an upper estimate for a randomly chosen pair, not a typical one. That is the favourable reading for multiplexing: the per-gene effect a screen must resolve does not shrink as k rises, so eq. (6) may treat $|\Delta|$ as fixed in k. The comparison is made *within* Sameith — same lab, platform and noise floor — because comparing Kemmeren's singles against Sameith's doubles would confound perturbation count with study.

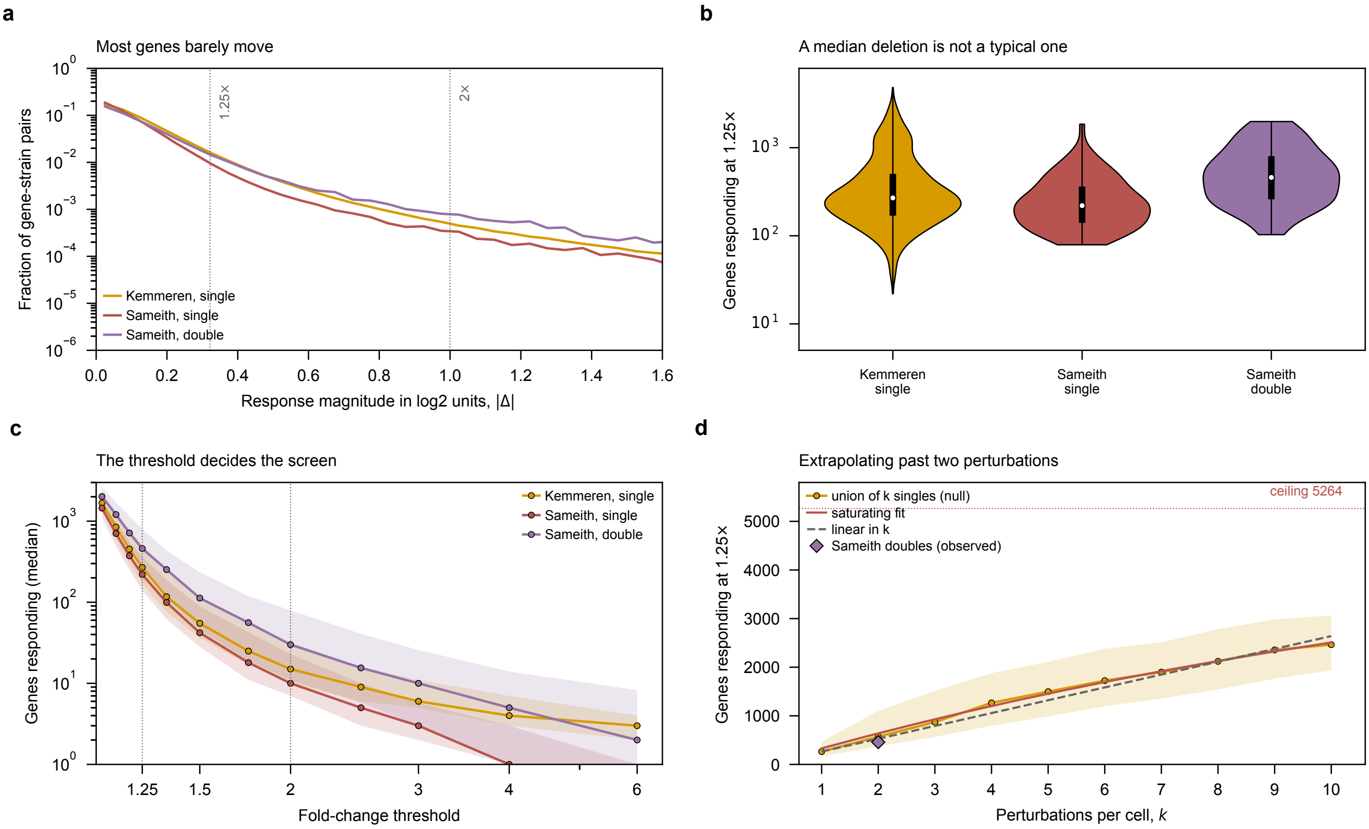

Figure 5: What differential expression after a gene deletion actually looks like, and why a two-fold design target is the wrong one for this biology. All three panels are the Kemmeren and Sameith compendia, singles and doubles shown separately. (a) Distribution of response magnitude over every gene-in-every-strain pair. It is a steep decay: the mass sits below $|\Delta| = 0.5$, and the two thresholds the document argues about fall far out on a thinning tail, so the threshold chosen decides how much of the response is visible at all rather than merely how it is filtered. The double-deletion curve sits above the singles across the whole range, but the shape is the same — more perturbation moves more genes, it does not change the character of the response. (b) Responders per strain at 1.25×, as a full distribution. Violin width is a kernel density over $\log_{10}$ counts — these distributions are strongly right-skewed as counts (skew 1.3–3.3) but nearly symmetric once logged (0.08–0.56), so log space is where a density is a fair summary; the black bar is the interquartile range, the thin line the observed full range, and the white dot the median. The interquartile spread *within* each group is wider than the gap between singles and doubles, and the range runs from 22 genes to 4,810 — a factor of 18 above the median at the top end, with 12% of Kemmeren's strains moving more than a thousand genes. A screen is not designed for the median deletion, because the median deletion is not a typical one, and cells-per-perturbation has to be set for the quiet end. (c) The design curve — median responders against the threshold, shaded to the interquartile range. Between 1.25× and 2× the count falls by roughly 15–20-fold in every dataset. That collapse is the cost of a two-fold target stated in the currency that matters: choosing it discards about 95% of the genes that actually responded. (d) Extrapolating past two perturbations. Sameith gives only $k = 1$ and $k = 2$, and any two points are fitted exactly by a line, a power law or a saturating curve, so a fit to them cannot choose a model. Instead each of Kemmeren's 1,484 single-deletion responder *sets* is known, so drawing k at random and taking their union measures directly what k combined perturbations would move if their effects did not interact — with the real overlap structure, which matters because stress and ribosomal genes respond to almost anything. Shading is the interquartile range over 2,000 draws. The saturating form $R(k) = G_{\text{eff}}(1 - (1 - p)^k)$ fits with $G_{\text{eff}} = 5{,}264$ and $p = 0.063$; a purely linear extrapolation is shown for comparison and the two are indistinguishable to $k \approx 8$, diverging only as the ceiling begins to bind. Sameith's observed doubles (diamond) sit 8% above the same null built from Sameith's own singles, which is the size of the epistatic correction at $k = 2$. Generated by `experiments/019-perturb-seq-costing/scripts/plot_effect_size.py`.

The design target is 1.34-fold, and it is expensive. Every cells-per-perturbation number derived under a two-fold assumption understates the requirement by $\sim$5.6$\times$. What this does *not* do is invalidate the 100-cell convention — the opposite, as section 4.7 shows: at the measured effect size that convention corresponds to an ordinary biological overdispersion rather than an implausible one.

What is still missing. These compendia are bulk microarray, so they measure the *effect size* $|\Delta|$ and say nothing about φ_j, the between-cell variance, which needs single-cell data. Of the three unmeasured inputs to eq. (5), this settles one.

4.5 Multiplexing and Interaction Coverage ✗

The end goal is multiplexed screens, because engineered strains carry stacked perturbations. The arithmetic splits cleanly into a part that helps and a part that does not.

Main effects get k times cheaper. Under an additive model with random k-subsets of T genes and N cells total, the cells informing gene g's main effect number Nk/T — independent of how many distinct combinations the library contains. Every cell carrying g contributes, whatever else it carries. So k-plexing divides the cell requirement by k.

Named interactions scale far worse. The probability that a specific pair $\{g, h\}$ both appear in a random k-subset is $k(k-1)/\left(T(T-1)\right)$. Requiring c cells on every pair among T targets therefore needs

$$N = c\binom{T}{2}\Big/\binom{k}{2}, \tag{3}$$

pairs needed over pairs delivered per cell. Yao et al. [11] derive the same expression and, importantly, supply the constant: $c = 400$, being "the number of cells needed to learn a single second-order interaction effect with the same power as our conventional screens had to learn first-order effects ($\sim$100 cells/perturbations), assuming that four times as many cells are needed to estimate a second-order interaction effect as a first-order effect." Our implementation of Eq. 3 reproduces their published worked cases to within rounding — 100,000 cells at $k = 3$ powers all pairs among 39.2 targets against their stated $\sim$39, and 10^6 cells at $k = 8$ gives 374.7 against their 374 — which is the check that the constant and the combinatorics are being applied correctly.

Table 9: What multiplexing buys, and what it does not. **Plex** k = guide RNAs delivered per cell, so k target genes are knocked down in that cell. **Main effects** = the average effect of one gene, pooling every cell that carries a guide against it whatever else it carries. **Pairs** are pairs of *target genes*, not plasmids: a cell informs a pair only if it happens to carry guides against both. Column 2 targets 250-cells-per-gene precision over 6,000 genes; columns 3–4 apply Eq. 3 at Yao's 400 cells per pair; column 5 inverts it – how many target genes can have *all* their pairs powered inside one 300,000-cell run.

Guides per cell (k)	Cells for main effects	Cells for all pairs among 6,000 genes	Cells for all pairs among 200 genes	Genes fully paired in 300k cells
1	1,500,000	–	–	–
2	750,000	7,198,800,000	7,960,000	39
3	500,000	2,399,600,000	2,653,333	68
5	300,000	719,880,000	796,000	123
8	187,500	257,100,000	284,286	205
10	150,000	159,973,333	176,889	260

The design that follows. Genome-wide pairwise epistasis is out of reach at any plex: 2.6×10^9 cells at $k = 3$, still 2.6×10^8 at $k = 8$. But a *focused* library inverts the picture, because Eq. 3 is quadratic in T and quadratic in k. A 200-gene metabolic sub-library at $k = 8$ needs $\sim$284,000 cells for *every* one of its 19,900 pairs — one genome-scale-sized run. That is the single most useful number in this document for the multiplexed screening goal.

So the programme splits cleanly: run genome-wide at $k \approx 3$ for main effects, where compressed decoding applies and interactions are assumed to cancel; and run a focused metabolic sub-library at high plex for named epistasis, where they are the signal.

Three costs the cell arithmetic hides. Guide detection compounds: assigning k guides at per-guide detection q succeeds at q^k, and the two published rates bracket a very wide range. At Brandner et al.'s 25% in bacteria, even after enrichment, a 3-plex assigns 1.6% of cells and an 8-plex is arithmetically hopeless; at the >71% Nadal-Ribelles et al. reach in yeast, the same figures are 36% and 6%. This — not sequencing — is what actually caps plex for us, and which end of that range applies is the single most valuable thing the pilot can measure (Sec. 3.5). Barcode collisions change character: in a singles screen a collision attenuates a main effect slightly, but in a multiplex screen it *manufactures a* *gene pair that was never constructed*, and pairs are precisely what is being measured. And Yao et al.'s own screens, despite being designed for this, recovered no significant pairwise interaction even at $q < 0.5$ — a caution that the 400-cell constant is a power calculation, not a demonstration.

4.6 Library Construction Limits ✗

Multiplexing is limited by cloning and transformation before it is limited by sequencing, and the binding number comes from our own library's paper.

Lian et al. [13] put yeast transformation at 10^6–10^7 independent clones and set their own construction standard at $> 10^6$ clones with *at least 30-fold redundancy.* Those two numbers decide the question. A dual-guide library over 6,000 genes has $\binom{6000}{2} = 1.8 \times 10^7$ pairs, which at 30-fold redundancy needs 5.4×10^8 independent clones — 54 to 540 times beyond the ceiling. Genome-scale pairwise coverage is not merely expensive to construct; it is unreachable by about two orders of magnitude, and no sequencing budget changes that.

The precedent already ran this experiment. sMAGIC is the combinatorial version of our own library, and it hit exactly this wall: "due to the limitations in transformation efficiency ($\sim 10^6$–10^7 for yeast transformation), only $\sim$0.01% to $\sim$0.1% of the total combinations ($\sim 10^5 \times 10^5 = 10^{10}$) were covered." That is the same arithmetic seen from the other side, and it carries a design lesson: they reached those combinations by *co-transforming* single-guide libraries rather than by cloning pairs. Co-transformation converts a construction problem into a Poisson sampling problem — diversity becomes free, but which pairs you get is random, so coverage of a *named* pair is governed by eq. (3) rather than by the library design. Assembly is not the obstacle either way: Golden-Gate efficiency is "nearly 100%".

Restricting the panel is what makes it constructible. The same arithmetic that rules out the genome-scale library admits the focused one. A 200-gene metabolic sub-library has $\binom{200}{2} = 19{,}900$ pairs, needing 597,000 clones at 30-fold redundancy — comfortably inside a single transformation. This is the same conclusion section 4.5 reaches from cell counts, arrived at independently from cloning: panel size, not plex, is the variable that decides whether named epistasis is reachable, and it binds in both the wet lab and the budget.

4.7 A Unified Design Equation ✗

The preceding subsections give three rules that look independent: a shot-noise requirement (eq. (2)), a 100-cell biological floor, and a combinatorial penalty for multiplexing. Sections 4.3 and 4.5 combine the first two by taking whichever is larger. They are not independent, and the max is an artefact of having modeled the count as Poisson.

This is not a new result. It is the standard negative-binomial sample-size calculation from the bulk RNA-seq literature, applied to a pseudobulk pool. eq. (5) appears in essentially this form in Hart et al. [23]: a sum of a sampling term and a dispersion term over the squared log fold change. The parameter φ_j is the negative-binomial dispersion of McCarthy et al. [24], whose biological coefficient of variation is $\sqrt{\varphi_j}$ — so published BCV values transfer directly as priors for the one input we cannot measure — and the variance parameterization itself is that of Robinson and Smyth [25]. The decomposition into Poisson sampling on top of biological variation is the single-cell UMI noise model of Grün et al. [26], and using a plain negative binomial rather than a zero-inflated one follows Svensson [27]. Treating the estimand as a pseudobulk profile rather than a per-cell test follows Squair et al. [28]; the cells-versus-depth optimum that fig. 6a traces under a fixed budget is the subject of Zhang et al. [29]; and the largest published Perturb-seq screen, whose empirical cells-per-perturbation choice section 4.3 is measured against, is Replogle et al. [30]. What is specific to this document is the multiplexing term of eq. (6) and the reading of the 100-cell convention below.

Setup. Pool n cells carrying one perturbation, each read to depth d mRNA UMIs, and consider gene j with transcriptome share p_j. The pseudobulk count is $K = ndp_j$. Treating it as negative binomial, with φ_j the gene's biological overdispersion between cells — the variance in excess of sampling —

$$\mathrm{Var}(\log K) \;\approx\; \underbrace{\frac{1}{ndp_j}}_{\text{shot noise}} \;+\; \underbrace{\frac{\varphi_j}{n}}_{\text{biology}} . \tag{4}$$

Claim. Requiring a $\log_2$ fold change Δ to be resolvable against controls at level α and power $1-\beta$ gives one requirement,

$$n^* \;=\; A(\Delta)\left[\frac{1}{d\,p_j} + \varphi_j\right], \qquad A(\Delta) = \frac{\kappa\,(z_{1-\alpha/2} + z_{1-\beta})^2}{(\Delta \ln 2)^2}, \tag{5}$$

where $\kappa = 1$ when the non-targeting control pool is large enough that its own variance is negligible — the usual case, since controls are pooled over many guides — and $\kappa = 2$ for a control of the same size as the perturbation. and with k

guides per cell over a panel of T targets the total cells for an order-r effect are

$$N^* = \rho_r \, n^* \frac{\binom{T}{r}}{\binom{k}{r}} \cdot \frac{1}{s\,q^r}, \tag{6}$$

with ρ_r the contrast variance inflation, s the QC survival fraction and q the per-guide detection probability. ρ_1=1 by definition. ρ_2=4 is *quoted* from Yao et al. rather than derived: it is the variance of a full 2×2 factorial contrast over four equally sized groups. Under the same large-control assumption used for main effects the exact factor is 3, so Yao's 4 is conservative by about a third; it is kept because the 400-cells-per-pair constant of section 4.5 is built on it and the two must not disagree.

Consequence. eq. (5) contains both earlier rules as limits. Setting $\varphi_j \to 0$ leaves $n^* = A/(d\,p_j)$, that is a required pseudobulk count $K = n^* d\,p_j = A$; at $\kappa = 1$ this is *identically* eq. (2), so section 4.3 is the shot-noise corner of this equation rather than a separate rule. Letting $d\,p_j \to \infty$ gives $n^* \to A\varphi_j$, a floor no amount of sequencing can cross — which *is* the biological floor, now with a number attached. At two-fold and 80% power $A = 16.3$ at two-fold and $A = 90.9$ at the measured 1.34-fold, so a 100-cell floor is precisely the claim $\varphi_j \approx 6.1$ or $\varphi_j \approx 1.1$ respectively.

That number depends on Δ, and this is where measuring the effect size pays off. At the nominal two-fold the 100-cell floor implies $\varphi_j \approx 6.1$, a between-cell coefficient of variation of 2.5 — bursty even by single-cell standards, and grounds for suspecting the convention was really guarding against something other than sampling variance. At the *measured* 1.34-fold (section 4.4) the same floor implies $\varphi_j \approx 1.1$, a coefficient of variation of about one, which is an entirely ordinary single-cell overdispersion. The tension dissolves: the 100-cell convention is consistent with unremarkable yeast biology once it is evaluated at the effect size screens are actually built to detect. The convention was right; the two-fold assumption was wrong.

Table 10: Inputs to the design equation. Three of the seven are assumed rather than measured, and those three are what separate a structure from a predictor. **Status**: *measured* = from data we hold; *quoted* = stated by a source; *derived* = computed from measured values; **assumed** = a placeholder, and every number downstream of it inherits that status.

Symbol	Quantity	Value	Status	Source
$A(\Delta)$	power coefficient, Eq. (5)	16.3	**assumed**	Δ nominal two-fold; scales as Δ^{-2}
Δ	$\log_2$ fold change to resolve	1.0	**assumed**	measured by section 4.4
p_j	transcriptome share of target gene	1.2×10^{-4}	derived	3.5 molecules / 30,000 mRNA, table 5
φ_j	biological overdispersion between cells	0.5–10	**assumed**	swept; sets the floor. Not yet measured
d	sequencing depth, mRNA UMIs per cell	410–2,000	measured	per platform, table 6
ρ_2	variance inflation, second order	4	quoted	Yao et al.; the 400-cells-per-pair constant
q	per-guide detection probability	–	**assumed**	needs the construct of section 3.5

Interpretation — what would make this predictive. As a structure eqs. (4) to (6) are complete, and every design question in section 5 is a substitution into them. As a *predictor* they are not usable yet, because three inputs are assumed rather than measured, and they are the honest answer to what a real design equation would take:

1. **φ_j, gene-level biological overdispersion in yeast.** The critical one: it alone sets the floor, and the whole 100-cell convention rests on it. Estimable from any existing yeast scRNA-seq count matrix, and it is the reason the figure sweeps φ instead of fixing it.
2. **The distribution of Δ — now measured.** Median $|\Delta| = 0.42$ among responding genes, a $1.34\times$ change (section 4.4, table 8). $A(\Delta)$ goes as Δ^{-2}, so this was the most violent term in the equation; pinning it multiplied every cell requirement by ~5.6 and resolved the φ puzzle above. Two of the three remain open.
3. **q, per-guide detection probability.** Enters as q^r, so it is forgiving for main effects and punishing for interactions. It cannot be measured until the guide-barcode construct of section 3.5 exists.

Of the three, one is now measured and two remain. The curves here therefore rest on one measured parameter and one swept one, which is why φ is drawn as a family rather than a line, and why no number from this subsection is yet used in the budgets of section 5 — those still assume two-fold and are consequently optimistic by $\sim5.6\times$ in cells.

Two limitations of the derivation itself, for completeness. The delta-method step behind eq. (4) degrades when counts are small, and $A \approx 16$ counts is small — near the shot-noise limit eq. (5) should be read as an order-of-magnitude statement, with an exact negative-binomial test the right tool in that corner. And eq. (5) assumes equal per-cell depth; real libraries vary and are handled with size factors, so d should be read as a mean depth, correct to first order. Neither affects the structural conclusions, both of which concern the φ term. The published treatment of the cells-versus-depth optimum that panel (a) rediscovers is Zhang et al. [29], and the case for pseudobulk as the estimand is Squair et al. [28].

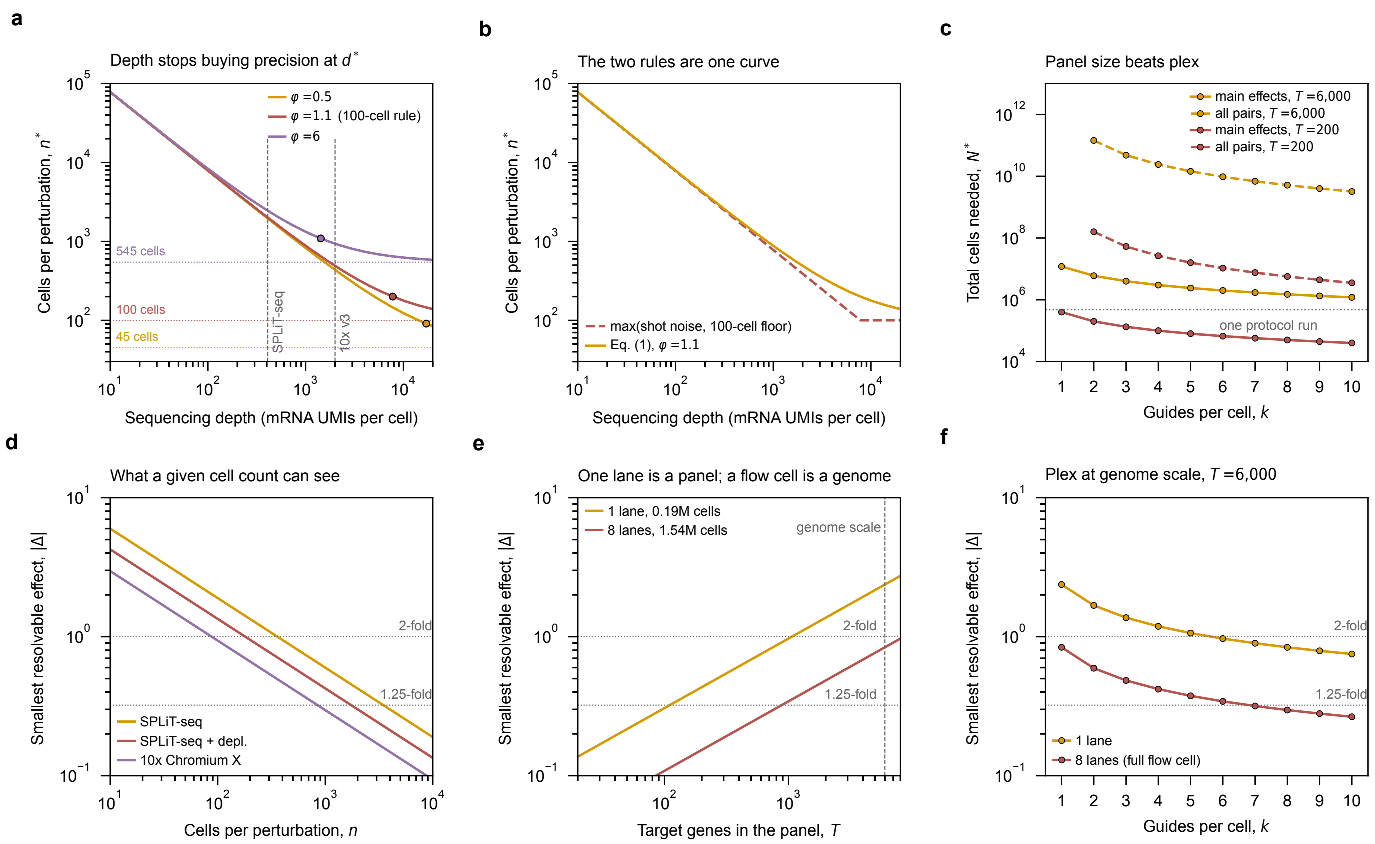

Figure 6: One equation reproduces both design rules, and turns a lane of sequencing into a statement about the smallest effect a screen can see. Top row, the structure of eq. (5). (a) Cells per perturbation against depth, swept over biological overdispersion φ. Every φ implies a cell floor, $n^* \to A\varphi$ as depth grows without bound, and those are the dotted horizontals — each labeled with the cell count it corresponds to, so the swept parameter can be read directly as the quantity a screen is designed in. The marked point on each curve is the depth-sufficiency threshold $d^* = 1/(\varphi p_j)$, where shot noise and biology contribute equally. Past d^* the requirement is within a factor of two of its floor, so further depth is close to wasted. Curves are evaluated at the *measured* effect size of section 4.4, $|\Delta| = 0.42$, not at a nominal two-fold; the middle curve is the $\varphi = 1.1$ that the field's 100-cell rule implies at that effect size, and its floor is accordingly 100 cells. At $\varphi = 1.1$, $d^* \approx 7{,}800$ UMIs per cell — roughly four times deeper than the best platform on offer — so *both* candidate chemistries are firmly shot-noise limited, and depth still buys precision for both. That is the argument for rRNA depletion in one number: it is the cheapest available way to move right along this axis. (b) The same curve against the piecewise rule the earlier subsections use. They agree exactly in both limits and differ near the knee, where the piecewise form *under*-estimates the requirement — the regime both candidate platforms occupy. (c) eq. (6) against plex, colored by panel size and dashed for pairwise. Raising k moves every curve down by a similar factor; shrinking the panel from 6,000 targets to 200 moves the pairwise curve by $\sim$900×. Restricting the panel, not multiplexing harder, is what brings interactions into reach. Bottom row, eq. (5) inverted — the smallest resolvable $|\Delta|$ — for the two sequencing configurations the core actually sells: one NovaSeq X 25B lane (3.2 billion read pairs, \$3,380) and a full eight-lane flow cell (25.6 billion, \$25,440 at the volume rate). (d) Resolution against cells per perturbation, by platform. (e) Against panel size, at split-pool with rRNA depletion. This is the panel to design from: one lane yields $\sim$192,000 usable cells and resolves two-fold across a few hundred targets, whereas a full flow cell yields $\sim$1.5 million and reaches two-fold across all 6,000 genes. One lane is a focused panel; a flow cell is a genome. (f) Against plex at genome scale — multiplexing to $k \approx 8$ buys a single lane roughly what eight lanes deliver unmultiplexed, which is the cheapest lever in the document. Note that the volume discount itself is minor (\$0.99 against \$1.06 per million read pairs, $\sim$6%); the reason to fill a flow cell is what it buys, not what it saves. Generated by `experiments/019-perturb-seq-costing/scripts/design_equation.py`.

5 Application: Genome-Scale Screen Design at UIUC ✗

Applying Sec. 4 to our actual instruments and rates. Design target: 6,000 genes, six guides per gene in the MAGIC style, one environment.

5.1 Start-Up Cost Composition ✗

Brettner et al. quote a start-up cost of ∼$10,000 and a per-protocol cost of $1,300–1,700. The composition matters more than the totals, because the two scale completely differently.

Table 11: Brettner 2024 SPLiT-seq line items (`paper.md:183`). The start-up cost is one purchase order — three plates of custom IDT oligos, good for 215 runs.

Item	Cost	Scaling
IDT barcode plates (RT r1 + ligation r2 + r3)	$7,699.40	**one-time, 215 runs**
Standalone oligos (lyophilized)	$270.60	one time
Maxima H Minus Reverse Transcriptase (ThermoFisher EP0753)	$840.00	**per run**
T4 DNA Ligase (NEB M0202M)	$270.00	**per run**
RNase inhibitors (SUPERase-In + Enzymatics)	$127.00	**per run**
Barcoding oligos drawn from the plates	$36.00	**per run**
Dynabeads MyOne Streptavidin C1	$14.00	per sublibrary
WGS fragmentation + ligation library prep (Enzymatics)	$39.00	per sublibrary

The start-up cost is a single purchase order: three 96-well plates of custom barcoded oligos from IDT, $7,699.40, which is 77% of it.

What "215 uses" actually means. The plates are *reagent stocks*, not devices that get reused. IDT ships them pre-diluted — 300 µL of 100 µM oligo in each of 96 wells — and a single pass through the protocol withdraws roughly one microlitre from every well. Nothing is recovered afterwards and no barcode is ever read twice; the plate simply has enough volume in it for about 215 withdrawals. Brettner et al. state the capacity per plate directly: "enough volume to complete 300 RT, 250 r2, and 215 r3 ligations with careful pipetting." The round-3 plate is the binding constraint at 215, so 215 is the honest number rather than the 300 the RT plate would allow.

Two practical points follow from the pre-diluted format, and they answer the obvious worry about committing a whole plate at once. Because the oligos arrive already in solution, there is no rehydration event — you are drawing from a reservoir, not reconstituting a plate that must then be consumed. What does degrade them is freeze–thaw, which is why the working practice is to aliquot each source plate into several *working plates* and cycle only those. Gaisser et al.'s troubleshooting guide treats "remake fresh working plates" as the standard response to a barcoding failure, which tells you the source plates are expected to outlive many working copies.

And what counts as one "run". One run is one complete pass through the barcoding workflow of Sec. 3.1: fix and permeabilize a batch of cells, distribute them across the round-1 plate, two ligation rounds with pooling and re-splitting between, then split into sublibraries and prepare libraries. It consumes one withdrawal from each barcode plate, one tube of reverse transcriptase, one tube of ligase, and about a fifth of the RNase inhibitors — and it can carry up to 96 samples and, at Brettner et al.'s loading, ∼480,000 cells. It is a unit of *bench work*, not a unit of sequencing: the sublibraries a run produces are sequenced separately and are priced separately.

So the plate set is a capital asset. At a realistic five to ten runs per year, 215 withdrawals is a twenty-year supply, and it will be retired by degradation long before depletion. Two consequences: we are over-buying by roughly an order of magnitude and should ask IDT for a smaller synthesis scale or volume, and one plate set can serve an entire department. The genuinely recurring items are the reverse transcriptase ($840) and the ligase ($270), repurchased every run.

Independently replicated. Gaisser et al. [7] costed the same chemistry for bacteria and put the identical item at $7,844.52 against Brettner et al.'s $7,699.40 — two labs, two organisms, two years apart, 1.9% apart. Their barcode cost per run is flat regardless of sample count, which is the split-pool economics in one number.

5.2 Loaded Cost per Usable Cell ✗

The figure the literature quotes is library preparation only.

That spread — about four orders of magnitude — is real but misleading, because every row excludes sequencing and every row counts cells *processed* rather than cells that survive QC and carry an identified guide. Costing the same platforms end to end:

Table 12: Cost per cell, *library preparation only.* "Quoted" means the source states a per-cell rate; "derived" means a total divided by a cell count the same source supplies. Every row excludes sequencing.

Method	Organism	Isolation	$ / cell	Basis	Source
MATQ-seq	*bacteria*	plate	$50.00	derived	Gaisser 2024, Nat Protoc, Suppl. Table 2
Plate FACS + STRT-seq (Nadal-Ribelles 2019)	*S. cerevisiae*	plate	$4.15	quoted	Jariani 2020, eLife 9:e55320
10x Chromium 3' v2 (Jariani 2020)	*S. cerevisiae*	droplet	$0.3000	quoted	Jariani 2020, eLife 9:e55320
ProBac-seq	*bacteria*	droplet	$0.1649	derived	Gaisser 2024, Nat Protoc, Suppl. Table 2
PETRI-seq	*bacteria*	split-pool	$0.0367	derived	Gaisser 2024, Nat Protoc, Suppl. Table 2
microSPLiT (1 sublibrary)	*bacteria*	split-pool	$0.0172	derived	Gaisser 2024, Nat Protoc, Suppl. Table 2
SPLiT-seq (Brettner 2024)	*S. cerevisiae*	split-pool	$0.00500	derived	Brettner 2024, Yeast 41:242
microSPLiT (full scale)	*bacteria*	split-pool	$0.00251	derived	Gaisser 2024, Nat Protoc, Suppl. Table 2

Table 13: Cost per cell end to end: 6,000 genes at 250 cells/gene, per cell that survives QC *and* carries an identified guide.

Platform	Reagents $/cell proc.	Loaded $/usable cell	Hidden mult.	Seq. share
SPLiT-seq, as published	$0.0049	**$0.1044**	21.3×	81.2%
SPLiT-seq + rRNA depl.	$0.0053	**$0.0383**	7.2×	44.2%
10x Chromium X	$0.1076	**$0.2424**	2.3×	19.2%

The headline inverts. On reagents, split-pool beats droplet by ∼22×. End to end it is 2.3×. The gap is rRNA and discarded cells — 94% of split-pool reads are ribosomal and ∼75% of sequenced cells are thrown away. With rRNA depletion the advantage returns to 6.3×, which makes depletion the single highest-leverage change available to us.

5.3 UIUC Core Sequencing Capacity ✗

Rates from the UIUC Roy J. Carver Biotechnology Center DNA Services, effective 2026-08-01, retrieved 2026-08-18. Listed rates are the on-campus rate; external users pay +30.8%.

Table 14: UIUC Carver Biotechnology Center sequencing rates, effective 2026-08-01, on-campus. External users pay +30.8%. The core has no NextSeq and no NovaSeq 6000.

Option	Instrument	$ / lane	Read pairs / lane	$ per M pairs
Nova X 25B PE150, 8+ lanes	NovaSeq X Plus	$3,180	3,200M	$0.99
Nova X 25B PE150, per lane	NovaSeq X Plus	$3,380	3,200M	$1.06
Nova X 10B PE150, 8+ lanes	NovaSeq X Plus	$1,710	1,200M	$1.43
Nova X 10B PE150, per lane	NovaSeq X Plus	$2,000	1,200M	$1.67
Nova X 1.5B SR100, per lane	NovaSeq X Plus	$1,360	800M	$1.70
Nova X 10B PE150, SPLIT LANE	NovaSeq X Plus	$1,160	600M	$1.93
Nova X 1.5B PE150, per lane	NovaSeq X Plus	$1,580	800M	$1.98
i100 25M SR100nt	MiSeq i100	$986	25M	$39.44
i100 25M 2x150nt	MiSeq i100	$1,450	25M	$58.00
i100 titration run (26nt + indexes only)	MiSeq i100	$398	5M	$79.60
i100 5M 2x150nt	MiSeq i100	$661	5M	$132.20

The available instruments. The core runs the NovaSeq X Plus and the MiSeq i100, plus PacBio Revio and Nanopore GridION. It has no NextSeq and no NovaSeq 6000. The microSPLiT protocol is written against a NextSeq 2000 P2, so that configuration is not available here; the choice is a ∼25 M-read MiSeq run or a ≥800 M-read NovaSeq X lane, with a 30-fold gap between them (fig. 7a). The NovaSeq X 25B is the highest-throughput configuration on campus at 3.2 billion read pairs per lane and 25.6 billion per flow cell, and at the 8+ lane rate it is also the cheapest per read, which is why every sequencing figure in this document is priced on it.

Both methods need paired-end reads, and it costs nothing. Neither family can use a single-read flow cell — with one read you get barcodes and no transcript, or transcript and no cell identity. They put the barcode on *opposite* reads, which is the thing to get right at submission:

Paired-end is free here because the largest flow cell is paired-end only and still wins on price per read: 25B PE150 at 8+ lanes is $0.99 per million read pairs against $1.06 for 10B SR100. That would not hold at a core whose cheapest option is single-read.

Three split-pool-specific hazards. PhiX at ≥10% is a protocol requirement, not hygiene — read 2 carries fixed linker sequence at defined cycles, so base diversity collapses and patterned flow cells mis-register clusters; combined with 70–90% valid-barcode rates only ∼72% of purchased reads become usable data, which is in the model. Unique dual indices

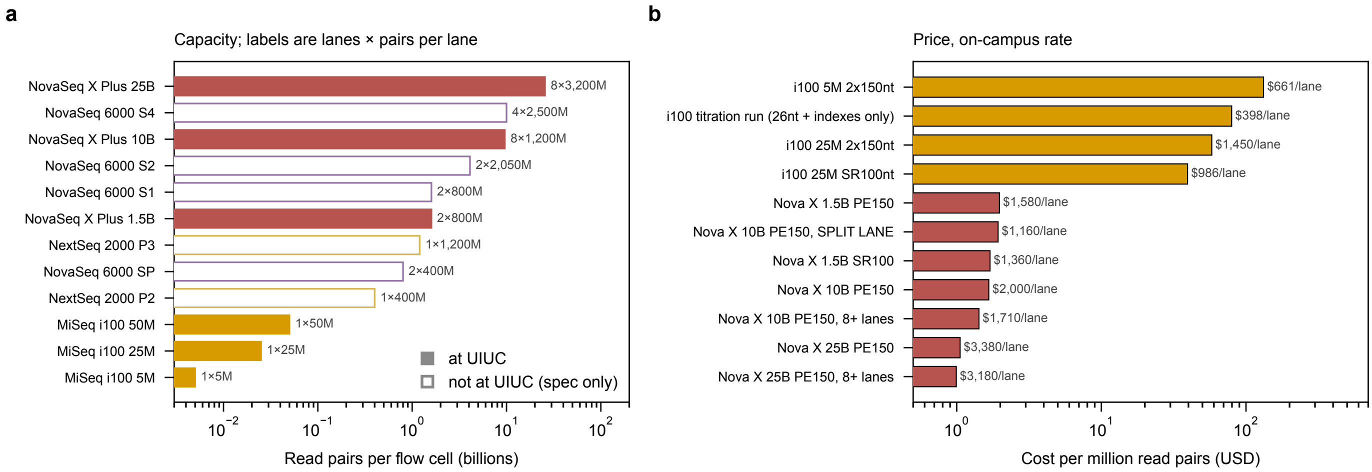

Figure 7: The instruments on campus leave a 30-fold hole exactly where a pilot screen wants to sit, and the largest flow cell is also the cheapest per read. (a) Output per flow cell, with lanes and reads-per-lane drawn separately, since a platform gets large by either route and the two are not interchangeable for a screen: lanes partition a library, reads-per-lane deepen it. Filled bars are available at the UIUC Roy J. Carver Biotechnology Center; open bars are manufacturer specifications for instruments the core does not have, shown for comparison and therefore unpriced. The gap between the MiSeq i100 at 50 M read pairs and the NovaSeq X 1.5B at 1.6 billion is the practical constraint — a pilot sized at a few hundred million reads has to either waste most of a NovaSeq lane or be redesigned down to a MiSeq, and the microSPLiT protocol's own NextSeq 2000 P2 configuration falls squarely in the hole. (b) Price per million read pairs, on-campus rate, for the configurations the core sells. Cost per read falls monotonically with flow-cell size, so the NovaSeq X 25B at 8+ lanes is simultaneously the largest and the cheapest option ($0.99 per million read pairs); there is no economy-of-scale penalty to buying the biggest flow cell, which is why every sequencing figure in this document is priced on it. A consequence worth noting for submission: the 25B is paired-end only, so the paired-end reads both method families require (Table 15) cost nothing extra here — a conclusion that would reverse at a core whose cheapest configuration is single-read. Generated by `experiments/019-perturb-seq-costing/scripts/plot_sequencers.py`.

Table 15: Read configurations. Both families require paired-end runs but put the cell barcode on *opposite* reads. For SPLiT-seq the index is barcode round 4, not merely a sample tag, so unique dual indices are mandatory.

Method	Read 1	Read 2	Index	Cycles
SPLiT-seq	74 nt cDNA	86 nt UMI+BC3+BC2+BC1	6 or dual 8 nt (= BC round 4)	160
10x Chromium 3'	28 nt barcode + UMI	90 nt cDNA	dual 10 nt	118
Drop-seq	20 nt barcode + UMI	100 nt cDNA	single 8 nt	120

are mandatory, because the sublibrary index *is* barcode round 4, so an index hop moves a cell into another sublibrary's namespace and manufactures a barcode collision rather than mere sample bleed. And a 2×150 kit gives 300 cycles when only 160 are needed, so read 1 should be set to 150 nt — free, and it helps with the whole-genome-duplication paralogs that forced Brettner et al. to keep multi-mapping reads.

5.4 Barcode Space and Collision Rates ✗

Table 16: Barcode collision rate, $100(1-((B-1)/B)^{C-1})$ with $B = 96^R \times S$. The sublibrary index S is a real barcode dimension: cells are split into sublibraries *after* the last in-cell ligation.

Rounds	Sublib.	Barcode space	@45k	@480k	@3M	@12M cells
3	1	884,736	4.96%	41.87%	96.63%	100.00%
3	24	21,233,664	0.21%	2.24%	13.18%	43.17%
3	96	84,934,656	0.05%	0.56%	3.47%	13.18%
4	1	84,934,656	0.05%	0.56%	3.47%	13.18%
4	**24**	**2,038,431,744**	0.00%	0.02%	0.15%	0.59%
4	96	8,153,726,976	0.00%	0.01%	0.04%	0.15%

What a fourth round buys, concretely. Each barcoding round multiplies the space by the number of wells, so three rounds of 96 give $96^3 = 8.8 \times 10^5$ trajectories and four give $96^4 = 8.5 \times 10^7$ — a **96-fold** increase, two orders of magnitude. Multiplied by 24 sublibrary indices that is 2.0×10^9 distinct labels against the ∼12 M cells a genome-scale screen needs.

The practical consequence is the sublibrary count. At 480,000 cells per run and a 1% collision target, three rounds need 54 sublibraries at $55 each in beads and library prep ($2,970 per run); four rounds clear 1% at a single sublibrary, so the count falls to whatever PCR complexity dictates — about 11 at Gaisser et al.'s 45,000-cells-per-sublibrary working limit, or $605 per run. A fourth plate costs ∼$2,566 once and ∼$150 per run, so it repays itself inside the first run.

Against that sits the one real argument: every split-pool ligation loses material, and the yield cost of a fourth round must be measured in the pilot before it is committed to.

5.5 Genome-Scale Budget ✗

Table 17: Genome-scale CRISPRi Perturb-seq budget: 6,000 target genes, six guides per gene, one environment, UIUC rates effective 2026-08-01. **Cells per target gene** = cells carrying a guide against that gene, pooled over all six of its guides – not cells per guide and not cells per plasmid. **Analysed** = cells surviving QC *and* assigned to a guide; **sequenced** = cells whose reads were paid for, which is larger because ∼75% of split-pool cells are discarded after sequencing. **Protocol runs** is a count of wet-lab repetitions of the barcoding workflow (split-pool) or of 10x channels, *not* a cost. All sequencing is priced on NovaSeq X 25B PE150 at the 8+ lane rate, the cheapest per-read option the core offers. Excludes labor, oligo pool synthesis, strain construction, and the ∼$10,000 one-time split-pool start-up.

Cells per target gene	Platform	Cells analyzed	Cells sequenced	Protocol runs	Read pairs (billions)	Reagents	Sequencing	Total
100	SPLiT-seq	600,000	2,400,000	5	50.0	$11,470	$50,880	**$62,350**
100	SPLiT-seq + depl.	600,000	2,400,000	5	10.0	$12,470	$12,720	**$25,190**
100	10x Chromium X	600,000	1,090,910	55	27.3	$117,865	$28,620	**$146,485**
250	SPLiT-seq	1,500,000	6,000,000	13	125.0	$29,470	$127,200	**$156,670**
250	SPLiT-seq + depl.	1,500,000	6,000,000	13	25.0	$32,070	$25,440	**$57,510**
250	10x Chromium X	1,500,000	2,727,273	137	68.2	$293,591	$69,960	**$363,551**
500	SPLiT-seq	3,000,000	12,000,000	25	250.0	$57,185	$251,220	**$308,405**
500	SPLiT-seq + depl.	3,000,000	12,000,000	25	50.0	$62,185	$50,880	**$113,065**
500	10x Chromium X	3,000,000	5,454,546	273	136.4	$585,039	$136,740	**$721,779**

Reading the table: the dominant term *moves*. For split-pool as published, sequencing is 81% of spend; for 10x it is 19% and library preparation is 81%, because a genome-scale screen needs 137 channels at $2,143 each. These are different businesses and the optimizations do not transfer between them. rRNA depletion is worth ∼$99,000 at the 250-cells-per-gene tier. And the 13 protocol runs behind the split-pool row are ∼13 weeks of skilled bench work with 13 chances to fail, against 137 channels the core runs for us — a real argument for droplet that the dollar column conceals.

Recommended entry point. 100 cells per gene, split-pool with rRNA depletion: ∼$25 k recurring plus ∼$10 k one-time. That is a genuine genome-scale yeast Perturb-seq for under $40 k of consumables and sequencing, and it is defensible because 100 cells per perturbation is the published field standard.

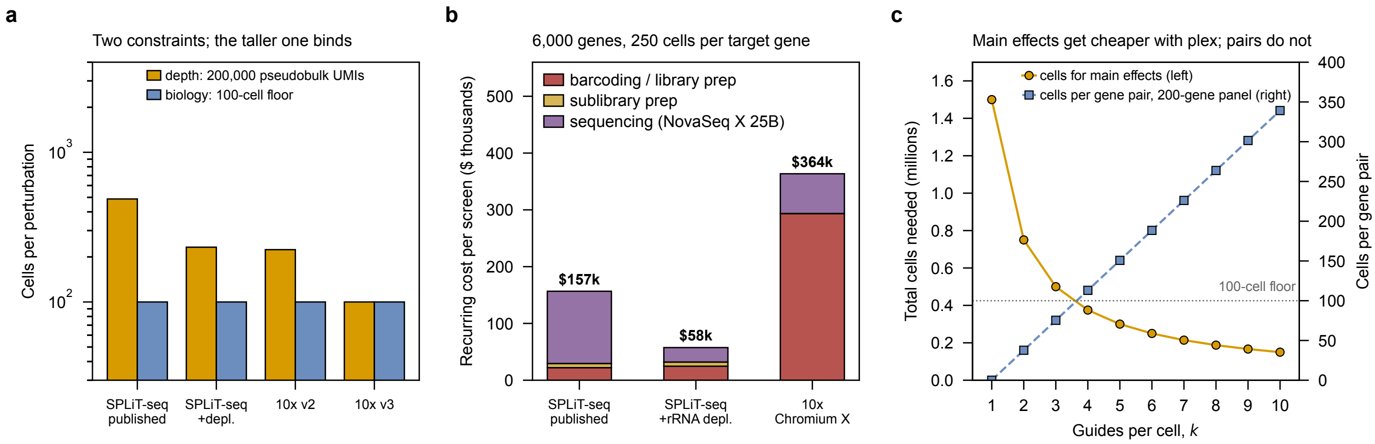

Figure 8: Cells per perturbation are set by biology rather than by sequencing at useful depths, the dominant cost term is different for the two platforms, and multiplexing makes main effects cheap without making pairwise interactions reachable. (a) Cells needed per perturbation under the two independent constraints of section 4.3: the depth requirement, i.e. how many cells it takes to accumulate 200,000 pseudobulk UMIs at each method's measured per-cell yield, and the 100-cell biological floor below which cell-cycle and metabolic state are not averaged out. The taller bar binds. The floor is the binding constraint for every deeper method and is within a factor of two of binding even for split-pool as published, so past a few hundred UMIs per cell the design is limited by how many cells must be observed, not by how deeply each is read — deeper chemistry stops buying anything, which is the argument for spending the improvement on cells instead. (b) Where the recurring money goes for a 6,000-gene screen at 250 cells per gene. The dominant term moves between platforms: sequencing is 81% of split-pool spend, while for 10x it is 19% and library preparation is 81%, because a genome-scale screen consumes 137 channels at $2,143 each. These are different cost structures, and an optimization that helps one (rRNA depletion, worth ∼$99,000 here) does nothing for the other. The three-fold spread between the cheapest and dearest column, $58 k against $364 k, is a platform decision, not a procurement one. (c) What multiplexing buys as guides per cell k rises. Left axis: total cells needed to hold main-effect precision across 6,000 genes falls as $1/k$, from 1.5 M cells at $k = 1$ to 0.15 M at $k = 10$, because every cell reports on k genes at once. Right axis: what that same cell budget delivers for one *named* gene pair if the library is restricted to a 200-gene panel, which grows roughly linearly in k and crosses the 100-cell floor near $k = 4$. The two curves are the whole design argument. Multiplexing alone does not open up epistasis — run the same calculation over all 6,000 genes and coverage per pair falls by $(200/6000)^2 \approx 1/900$, to well under one cell per pair at any k, because the pairs to be covered grow quadratically while the pairs a cell reports grow only as k^2. Restricting the panel is what makes pairs measurable, and multiplexing is what then makes it affordable; neither substitutes for the other. Generated by `experiments/019-perturb-seq-costing/scripts/plot_economics.py`.

6 Bottlenecks and Development Priorities ✗

What has to be true for 8–10 simultaneous perturbations to be measurable, and what to do first given that we will start with single-guide CRISPRi and move to two.

6.1 Priority Ranking ✗

1. Guide capture (blocking). Nothing else matters until guide identity is recoverable from a poly(A) readout (Sec. 3.5). It is a construction problem rather than a cost problem, and it now has a published yeast template. Let q be the *per-guide detection rate* — the probability that one guide present in a cell is actually seen in that cell's reads. A k-plex cell is only usable if all k of its guides are detected, so the usable fraction is q^k, and the whole feasibility of multiplexing turns on which q we inherit. At Brandner et al.'s bacterial $q = 0.25$ a 3-plex yields $0.25^3 \approx 1.6\%$ of cells and an 8-plex is arithmetically hopeless. At the $q > 0.71$ Nadal-Ribelles et al. achieve in yeast, a 3-plex yields 36% and even an 8-plex yields 6% [4]. That is the difference between multiplexing being impossible and merely expensive, and it is one measurement. Raising q is worth more than any other single change, because it enters the multiplex ceiling exponentially while everything else enters linearly — and the pilot's first job is to find out which of those two regimes a plasmid-borne guide barcode lands in.

2. mRNA fraction. At 5.75% mRNA, 94 of every 100 reads buy ribosomal RNA and the sequencing bill is inflated by roughly the reciprocal. Two independent levers: shift the random-hexamer:oligo-dT ratio, since the 5′ coverage hexamers buy is worthless for a 3′-counting screen; and add Cas9-guided rRNA depletion after cDNA synthesis. Brandner et al.'s depletion took *E. coli* from 4.6% to 60.2% mRNA, but only 2.2% to 10.4% in *P. putida*, so the transfer to yeast must be measured, not assumed. Worth ~\$99 k at genome scale (Sec. 5.5).

3. Cells per RT well. Brettner et al. verified 5,000 yeast cells per well; microSPLiT runs 40,000–100,000 for bacteria. This number sets how many cells one protocol run can barcode, and therefore how many runs a screen needs. The 250-cells-per-gene split-pool row of table 17 requires 13 protocol runs at Brettner et al.'s loading; at 20,000 cells per well the same screen is four runs, because a run would barcode ~1.9 M cells instead of ~480,000. That is ~13 weeks of skilled bench work reduced to ~4, which is most of the labor argument for droplet (Sec. 5.5). It is also the cheapest experiment on this list.

4. Barcode space. A fourth ligation round is necessary at genome scale and pays for itself within one run (Sec. 5.4). Its only cost is molecular yield through an extra ligation, which the pilot must measure. Under multiplexing the requirement tightens further, since a collision fabricates a gene pair rather than merely diluting a main effect.

5. Transformation efficiency. The ceiling on multiplexing is library construction, not measurement (Sec. 4.6). Combinatorial diversity generated by co-transforming several single-guide libraries, rather than by constructing combinations, is the route worth costing — it converts a combinatorial cloning problem into a Poisson loading problem.

6. Effect size. Sec. 4.4 is unresolved and everything in Sec. 4.3 inherits its uncertainty.

6.2 Staged Execution Plan ✗

1. **Construct and validate the poly(A) guide barcode** in an arrayed set of ~10 strains with known knockdown phenotypes, scored against bulk RNA-seq from the same strains. Ground truth is only available while the design is arrayed; a pool cannot tell you what was in a given cell.
2. **Establish the chemistry numbers** the model is most sensitive to: mRNA fraction after depletion, cells per RT well, guide-assignment rate q, the read-depth saturation curve, and — from the same data, at no extra cost — the transcriptome share p_j of a few representative target genes, which section 4.1 shows is currently worth a five-fold spread in cells per perturbation and is the largest unforced uncertainty in the budget. One \$398 MiSeq titration run protects a \$3,180 NovaSeq lane, and there is no earlier checkpoint — Brettner et al. is explicit that cDNA traces and library concentration do not predict barcoding success.
3. **Single-guide pilot** on a focused sub-library, at the 100-cell floor.
4. **Two-guide screen** on the focused metabolic sub-library, where named pairs are reachable, to measure how guide detection and collisions actually degrade at $k = 2$ before trusting any extrapolation to $k = 8$.
5. **Genome-scale single-guide screen**, four barcoding rounds, one environment.

6.3 Outlook ✗

Two extensions change what the platform is for, and both are worth designing toward now rather than retrofitting.

Splitting the sample — and why the obvious version does not work. Pairing a transcriptome with mass spectrometry on the *same* perturbation is the extension that would matter most for metabolic engineering, since flux is what we actually want and it cannot be inferred from expression alone. Fulcher et al. [31] is the precedent for parallel transcriptome and proteome from the same single cells. But the naive version — take half the contents of a compartment — is not available in either format, and for different reasons worth separating.

A semipermeable capsule cannot be aliquoted. It is a hydrogel shell with a pore cutoff around 30 nm: enzymes up to ∼160 kDa diffuse in, and that permeability is the entire point, since it is what lets reagents reach the cells without breaking the compartment [32]. There is no valve; the pores are a property of the material, not a gate that can be closed. And the same cutoff is fatal to the specific application: *small molecules leak.* Metabolites are precisely the species that pass a 30 nm pore freely, so an SPC cannot hold a metabolome in the first place, let alone divide one. Proteins are the more plausible target, since much of a proteome sits above the cutoff — but a metabolomic readout from SPCs is not a matter of engineering the split, it is excluded by the format.

Two routes remain, and they split the population rather than the compartment. Capsules are FACS-sortable and their cells can be released alive by brief protease treatment, so whole capsules can be *routed* — one fraction sequenced, another pooled for bulk mass spectrometry — which gives paired measurements over matched sub-populations rather than over identical cells. Alternatively, because each capsule holds a *clonal* population expanded from one founder, releasing that clone and dividing the cells does give two aliquots of the same genotype; what is lost is the capsule identity, which then has to be re-established by a barcode rather than by physical containment.

The unsolved part. Either route needs the mass-spectrometry channel to carry perturbation identity, and MS has no equivalent of a sequencing barcode. The plausible designs are chemical isobaric labeling of routed fractions, or physically arraying sorted capsules into wells so that position encodes identity — which trades throughput for it, and is the obvious tension to think about before this becomes a plan. Isotope labeling for flux sits downstream of solving that.

Environments as a second axis. The perturbation and environment axes are independent, so a screen is really a choice of subset of $\mathcal{L} \times \mathcal{E}$, and environments multiply cell demand linearly. Since split-pool uses the round-1 barcode for sample identity and a pooled screen needs only one sample, that plate's multiplexing capacity is free for environments — which is the axis that most enlarges the design space for model training.

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